TUAB data preparation for LaBraM — from raw EDF to model input

📄 **PDF version of this notebook** (figures, tables and formulas, no code; regenerate with section 13).

This notebook walks through, step by step and on **real recordings**, how the Temple University Hospital **Abnormal EEG Corpus (TUAB v3.0.0)** is turned into the tensors LaBraM is fine-tuned on. Every step is (1) explained, (2) written as a numbered formula, and (3) visualised on a **normal** and an **abnormal** recording.

The code that does this in the repo is `dataset_maker/make_TUAB.py` (offline, EDF → pickle) plus a few lines at training time in `labram/train/train_finetune.py` (pickle → model input). The notebook **re-implements the pipeline with the same MNE calls and the same constants (imported from `make_TUAB.py`)** and checks in section 7 (Table 7.1) that it reproduces the stored pickles exactly. Table 0.1 lists the steps.

Numbering. Equations, tables and figures are numbered per section (`Eq. (4.3)`, `Table 2.2`, `Figure 4.2(b)`); every table and figure sits under its own `Table s.k.` / `Figure s.k.` heading, and every panel carries a letter.

Table 0.1. The pipeline at a glance

Shapes are `[channels, samples]`; T is the recording length in seconds.

#	Step	Where	Shape after the step	Section
0	Read EDF, convert to µV	`mne.io.read_raw_edf`	`[C_raw ≈ 30–36, 250·T]` @ 250 Hz	2
1	Drop non-EEG channels, reorder to 23 standard channels	`raw.drop_channels`, `raw.reorder_channels`	`[23, 250·T]`	3
2	Band-pass FIR 0.1–75 Hz (zero phase)	`raw.filter(0.1, 75)`	`[23, 250·T]`	4
3	Notch FIR at 50 Hz	`raw.notch_filter(50)`	`[23, 250·T]`	5
4	Resample 250 → 200 Hz (FFT)	`raw.resample(200)`	`[23, 200·T]`	6
5	Cut into non-overlapping 10 s windows, label = folder	`X[:, 2000k : 2000(k+1)]`	`K × [23, 2000]`	7
6	*(training time)* scale ÷100, split into 1 s patches	`train_finetune.py`	`[23, 10, 200]`	8
7	*(optional, LaBraM++)* common average reference + per-patch z-score	`labram/data/preprocess.py`	`[23, 10, 200]`	9

Contents

1. The TUAB corpus: layout, labels, splits
2. Raw EDF signals, and the grand-average spectrum of the cohort (50 / 60 / 75 Hz peaks)
3. Step 1 — channel selection & ordering
4. Step 2 — band-pass filter 0.1–75 Hz (30 s before/after views)
5. Step 3 — 50 Hz notch filter (and why 60 Hz survives)
6. Step 4 — resampling 250 → 200 Hz
7. Step 5 — windowing into 10 s samples, labels, verification against stored pickles

8. Step 6 — what the model actually receives (÷100, patches, channel indices)
9. Step 7 — optional LaBraM++ preprocessing (CAR + z-score)
10. Normal vs abnormal: gallery of examples & class-level spectra
11. Summary of formulas & caveats
12. Comparison with other state-of-the-art preprocessing pipelines (foundation models & brain-age models)
13. Export to PDF

> **Requirements:** `mne`, `numpy`, `scipy`, `pandas`, `matplotlib`, `torch` (all in `requirements.txt`). Point `TUAB_ROOT` (below, or the `TUAB_ROOT` environment variable) to the `v3.0.0/edf` directory. The `processed/` sub-folder produced by `make_TUAB.py` is needed for sections 2.1, 7, 10 and 11; those parts are skipped if it is missing.

Setup

```
MNE 1.4.2 | TUAB root: /data/datasets/EEG-public/TAUB/TUH_Abnormal/v3.0.0/edf
```

1. The TUAB corpus: layout, labels, splits

TUAB is a subset of the TUH EEG Corpus: routine clinical EEGs, each recording annotated by a neurologist as **normal** or **abnormal** (based on the clinical report). The label is a property of the **whole recording** — there is no time-resolved annotation.

```
v3.0.0/edf/
├── train/ ─┬─ normal/01_tcp_ar/aaaaaaav_s004_t000.edf
│           └─ abnormal/01_tcp_ar/...
└── eval/  ─┬─ normal/01_tcp_ar/...
            └─ abnormal/01_tcp_ar/...
```

- File name = `<subject>_<session>_<token>.edf` — e.g. `aaaaaaav_s004_t000` is subject `aaaaaaav`, session 4, the first EDF file of that session.
- **Label** comes only from the folder name: $y = 1$ for `abnormal/`, $y = 0$ for `normal/`.
- `01_tcp_ar` = recordings made with an *averaged reference* (AR) electrode configuration; every channel is a referential channel named `EEG XX-REF`.

How `make_TUAB.py` splits the data (subject-level, done separately for each class c):

$$\begin{aligned} \mathcal{S}^c &= \operatorname{shuffle}\big(\{\text{subject ids in } \texttt{train}/c\}\big), \\ \mathcal{S}^c_{\text{train}} &= \mathcal{S}^c_{[\,0:\lfloor 0.8\,|\mathcal{S}^c|\rfloor\,)}, \qquad \mathcal{S}^c_{\text{val}} = \mathcal{S}^c_{[\,\lfloor 0.8\,|\mathcal{S}^c|\rfloor:\,]} \end{aligned} \tag{1.1}$$

where $c \in \{\text{normal}, \text{abnormal}\}$ and $|\mathcal{S}^c|$ is the number of subjects; the official `eval/` folder becomes the **test** set. Table 1.1 and Figure 1.1 count recordings and subjects per folder. (Caveats about Eq. (1.1) are in section 11, Table 11.2.)

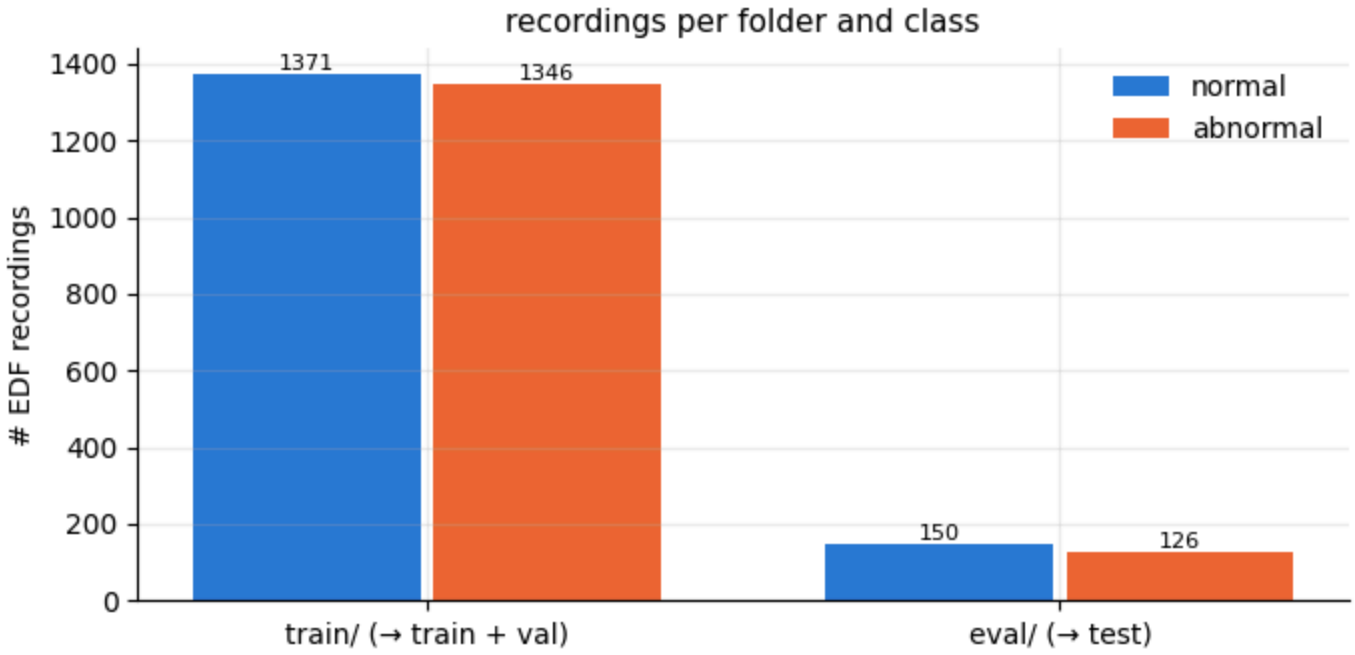
Table 1.1. Recordings and subjects per TUAB folder

`#recordings` counts EDF files, `#subjects` distinct subject ids (first token of the file name).

folder	y	recordings	subjects
train/normal	0	1371	1237
train/abnormal	1	1346	893
eval/normal	0	150	148
eval/abnormal	1	126	105

Figure 1.1. TUAB v3.0.0 recordings per folder

Number of EDF recordings per TUAB folder and class (all of TUAB v3.0.0; `train/` is later split into train + val by Eq. (1.1), `eval/` is the test set).



Examples used in the walk-through

We use recordings from the `eval/` (= test) folder, taken in sorted order so the choice is deterministic. Change `EXAMPLE_NORMAL` / `EXAMPLE_ABNORMAL` to look at other recordings. `SHOW_T0` is the start (in seconds) of the **10 s window that is shown after every processing step** (Figures 2.1, 3.2, 4.2–4.4, 5.2, 6.2, 7.2, 8.1, 9.1, 11.1). It is a multiple of 10 s, so the excerpt is exactly one of the stored windows of Eq. (7.1). The first minute of a clinical EEG often contains set-up artefacts, so we look a bit later.

```
normal    aaaaaayx_s002_t000: processed into 120 windows
abnormal  aaaaabdo_s003_t000: processed into 121 windows
```

2. Raw EDF signals

An EDF file stores each channel as 16-bit integers $d[n]$ together with a calibration in the header. MNE converts them to volts, and `get_data(units="uV")` converts to microvolts:

$$x[n] = 10^6 \left(p_{\min} + \left(d[n] - d_{\min} \right) \frac{p_{\max} - p_{\min}}{d_{\max} - d_{\min}} \right) \tag{2.1}$$

where $x[n]$ is the sample in µV, $d[n]$ the stored integer, and $d_{\min}, d_{\max}$ / $p_{\min}, p_{\max}$ the digital / physical (volt) range of the channel from the EDF header.

What the raw recordings look like (Table 2.1, Figure 2.1):

- sampling rate $f_s = 250$ Hz (anti-aliasing low-pass of the amplifier: 125 Hz),
- 30–36 channels — the 23 scalp/ear EEG electrodes **plus** non-EEG channels (EKG, EMG, eye leads `ROC/LOC`, photic stimulator, `IBI/BURSTS/SUPPR` analysis channels, unused numbered inputs `EEG 26-REF`, …),
- DC offsets and slow drifts, and power-line interference.

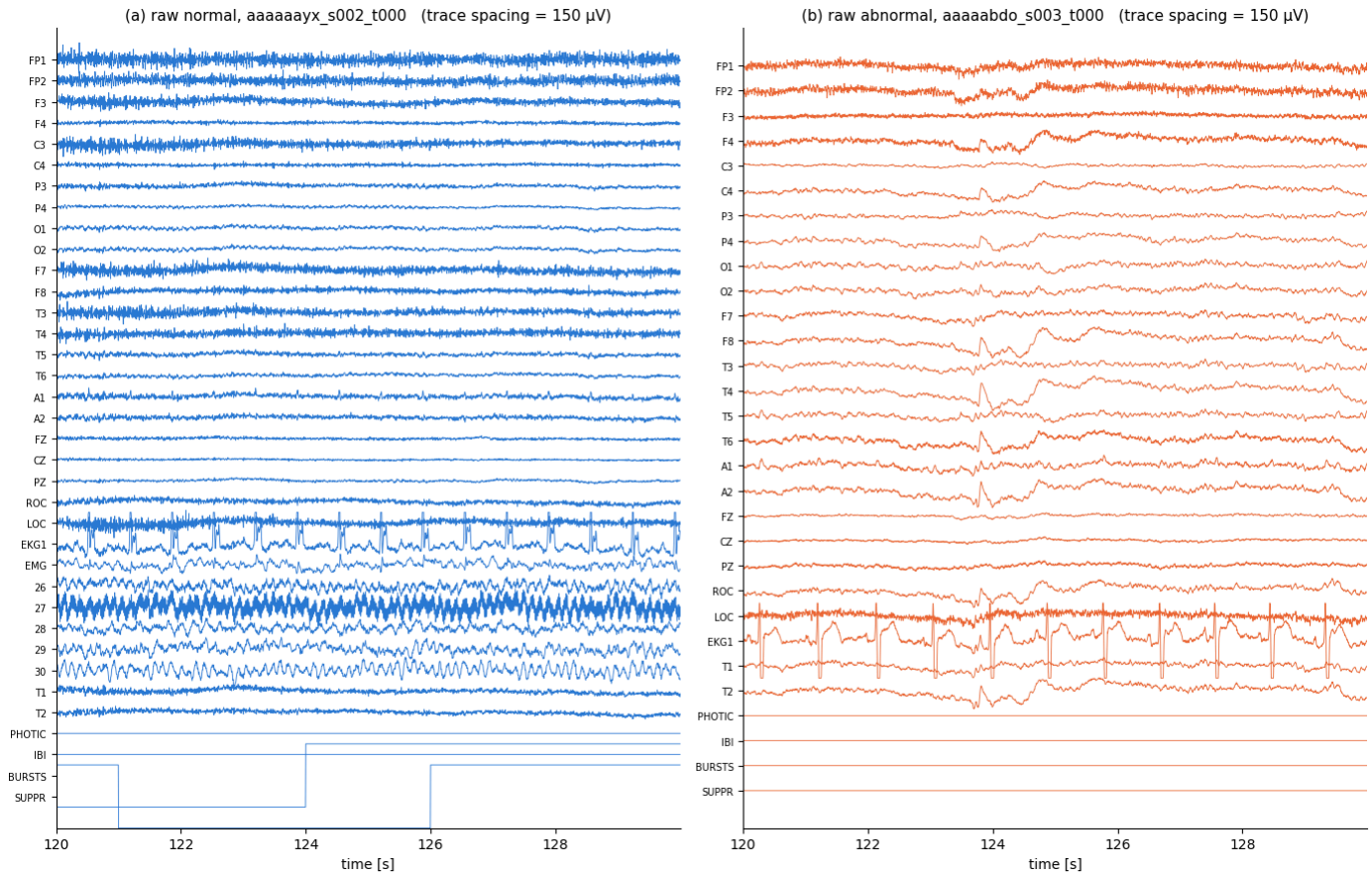
Table 2.1. Header of the two example recordings

`non-EEG channels` are the channels removed in step 1 (section 3).

class	recording	sfreq [Hz]	duration [s]	#channels	non-EEG channels
normal	aaaaaayx_s002_t000	250	1202	36	ROC, LOC, EKG1, EMG, 26, 27, 28, 29, 30, PHOTIC, IBI, BURSTS, SUPPR
abnormal	aaaaabdo_s003_t000	250	1217	30	ROC, LOC, EKG1, PHOTIC, IBI, BURSTS, SUPPR

Figure 2.1. Raw EDF, all channels (step 0)

Raw EDF, all channels, of (a) the normal and (b) the abnormal example, 10 s from `SHOW_T0`, 250 Hz, µV. Per-channel mean removed for display only; traces clipped at ±1.5 × spacing. The bottom rows (EKG, EMG, PHOTIC, IBI/BURSTS/SUPPR, numbered inputs) are not EEG.



The spectrum is estimated with **Welch's method** (`scipy.signal.welch` , helper `welch_psd`): the channel signal is cut into M Hann-windowed segments of L samples with 50 % overlap,

$$S_c(f) = \frac{2}{M\, f_s \sum_n w[n]^2} \sum_{m=1}^{M} \left| \sum_{n=0}^{L-1} w[n] \left(x_c[n + mL/2] - \bar{x}_{c,m} \right) e^{-j2\pi f n / f_s} \right|^2 \qquad (2.2)$$

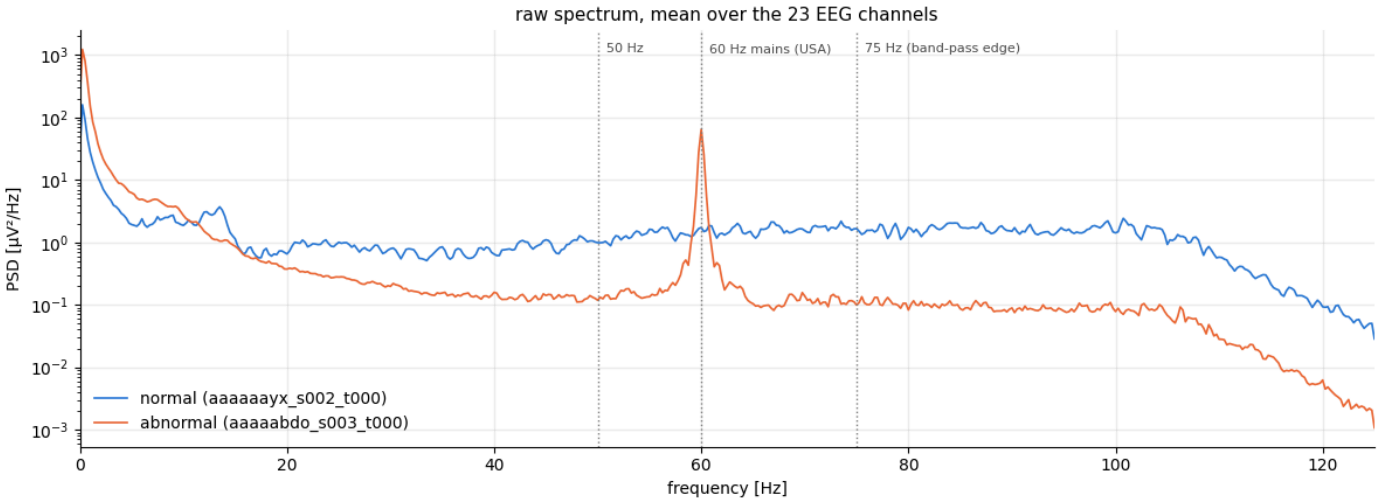
where $S_c(f)$ is the one-sided PSD of channel c in µV²/Hz, w the Hann window, $\bar{x}_{c,m}$ the segment mean (constant detrend), and $L = 4 f_s$ (4 s, 0.25 Hz resolution) unless stated otherwise. The **channel-averaged PSD** of a recording is

$$S(f) = \frac{1}{C} \sum_{c=1}^{C} S_c(f) \qquad (2.3)$$

over its $C = 23$ EEG channels. Figure 2.2 shows Eq. (2.3) for the two examples: a lot of power near 0 Hz (drift / DC), content all the way to the 125 Hz amplifier cut-off, and — very clearly in the abnormal example — a sharp line at **60 Hz** (US power mains). Section 2.1 checks how typical this is.

Figure 2.2. Raw spectrum of the two examples

Raw channel-averaged PSD, Eq. (2.3), of the two example recordings (23 EEG channels, whole recording, 250 Hz).



2.1 Grand-average PSD of the cohort and the 50 / 60 / 75 Hz peaks

Two recordings do not tell whether a spectral line is typical. We compute Eq. (2.3) for **every recording of the test set** (`eval/` , both classes), once on the **raw** signal (step 1: 23 EEG channels, native rate, unfiltered) and once on the **processed** windows stored by `make_TUAB.py` (step 5: after band-pass, notch and resampling; up to `GA_WINDOWS` evenly spaced 10 s windows per recording). Every recording PSD is interpolated onto a common 0.25 Hz grid. The **grand average** of a group $\mathcal{R}$ of recordings is

$$\bar{S}_{\mathcal{R}}(f) = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} S_r(f) \tag{2.4}$$

where $S_r(f)$ is Eq. (2.3) of recording r (µV²/Hz). Three statistics describe a spectral line at $f_0 \in \{50, 60, 75\}$ Hz in one recording (helpers `line_prominence_db` , `line_power_db`):

$$\Delta_r(f_0) = 10 \log_{10} \frac{\max_{|f-f_0| \le 0.5} S_r(f)}{\operatorname{median}_{1<|f-f_0|<5} S_r(f)} \tag{2.5}$$

$$\rho_r(f_0) = 10 \log_{10} \frac{\int_{f_0-0.5}^{f_0+0.5} S_r(f)\, df}{\int_{1}^{45} S_r(f)\, df} \tag{2.6}$$

$$\pi(f_0) = \frac{100}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} 1\big[\Delta_r(f_0) > \tau\big], \qquad \tau = 3 \text{ dB} \tag{2.7}$$

Δ_r (dB) is the **prominence**: how far the line sticks out of its own neighbourhood (0 dB = no peak, negative = a notch). ρ_r (dB) is the **line power relative to the 1–45 Hz EEG power** (-10 dB = the line carries a tenth of the EEG power). π (%) is the share of recordings with a visible peak. Figure 2.3 shows Eq. (2.4), Figure 2.4 the distribution of Eq. (2.5), and Table 2.2 summarises Eqs. (2.5)–(2.7).

```
recordings: 276 (normal 150, abnormal 126); with processed windows: 276
native sampling rates: {250.0: 268, 256.0: 8}
```

Figure 2.3. Grand-average PSD of the test set, raw vs processed

Grand-average PSD, Eq. (2.4), of all test recordings by class: (a) raw (step 1, 23 EEG channels, native rate), (b) zoom of (a) on 40–85 Hz with every recording's Eq. (2.3) as a thin line; (c) processed windows (step 5, 200 Hz), (d) zoom of (c). Shaded band = inter-quartile range across recordings; dotted lines at 50, 60 and 75 Hz. The grand mean (thick line) lies above the IQR band because a few high-amplitude (artefact-rich) recordings dominate the arithmetic mean of Eq. (2.4).

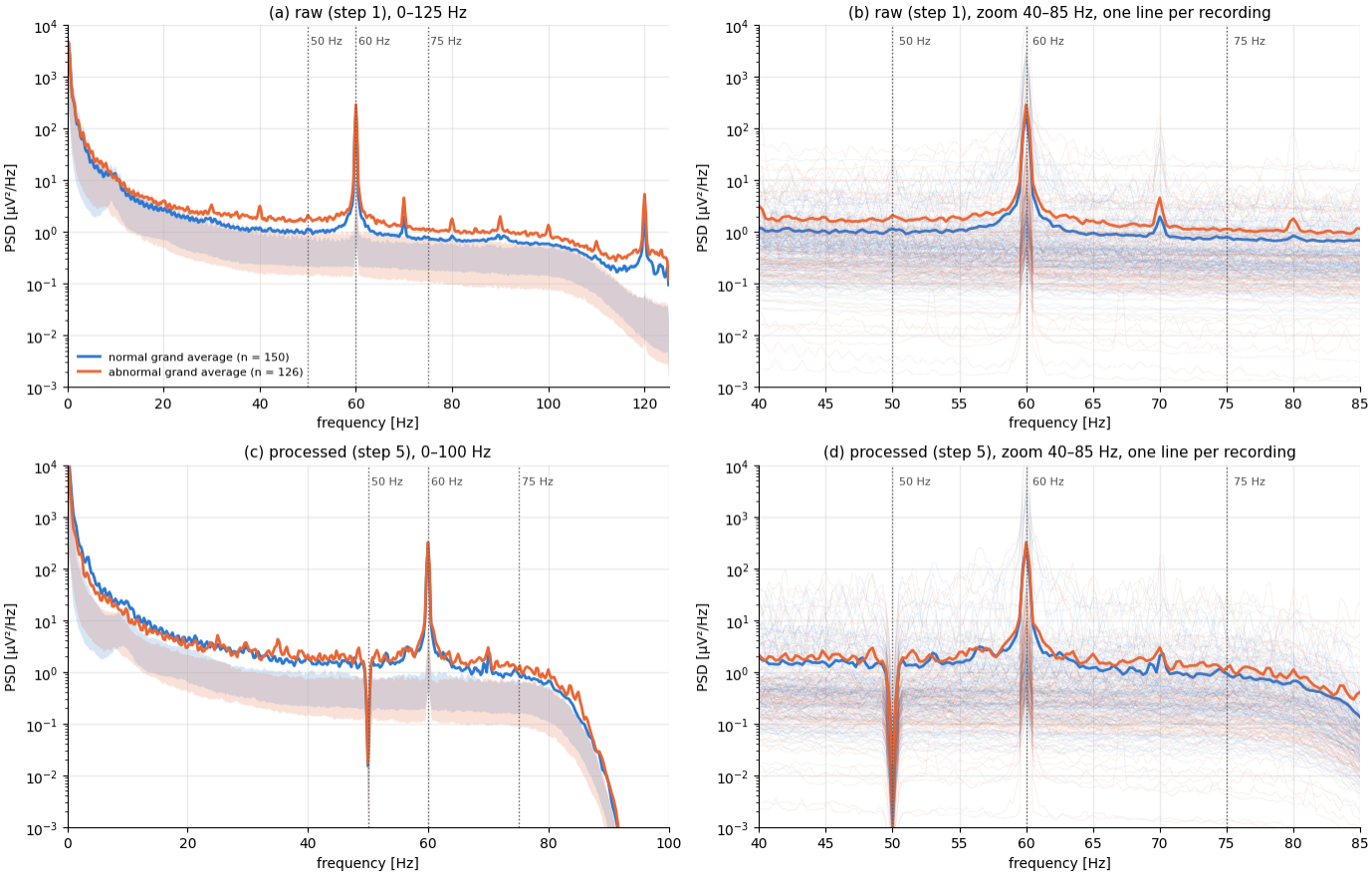


Figure 2.4. Line prominence at 50 / 60 / 75 Hz per test recording

Line prominence $\Delta_r(f_0)$, Eq. (2.5), per test recording at 50, 60 and 75 Hz, by class: (a) raw (step 1), (b) processed (step 5). Boxes = quartiles, whiskers = 5th–95th percentile, dots = recordings; dashed line = the $\tau = 3$ dB threshold of Eq. (2.7).

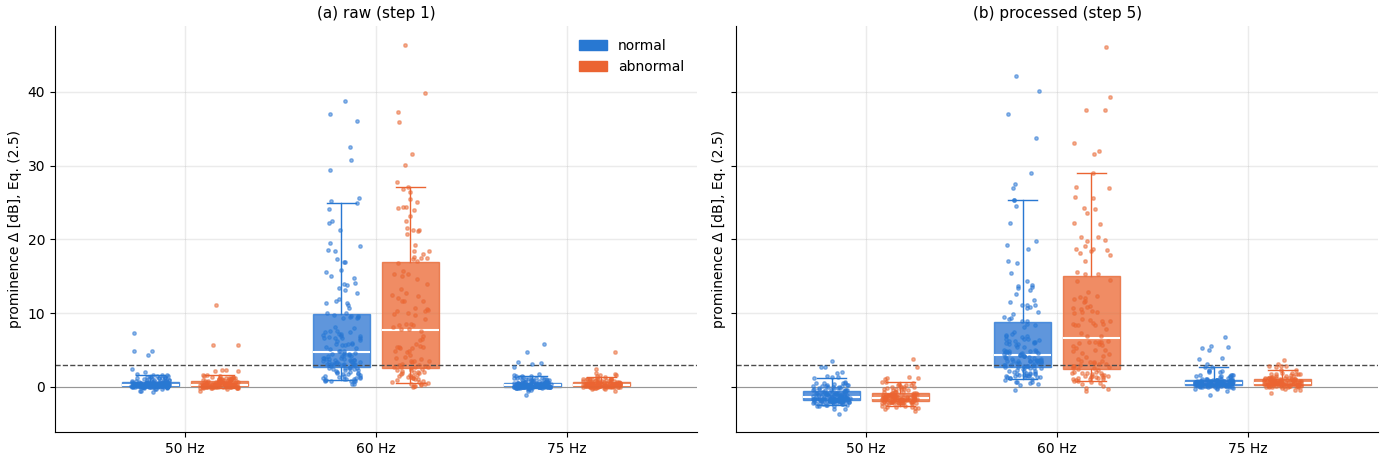


Table 2.2. Statistics of the 50, 60 and 75 Hz lines in the test set

One row per frequency and stage. `Δ` columns: median [inter-quartile range] of Eq. (2.5) in dB, per class and overall; `π` = Eq. (2.7), % of recordings with $\Delta > 3$ dB; `ρ` = median [IQR] of Eq. (2.6) in dB over all recordings.

line	stage	Δ normal [dB]	Δ abnormal [dB]	Δ all [dB]	π: Δ > 3 dB [%]	ρ all [dB]	n
50 Hz	**raw (step 1)**	0.3 [0.1, 0.7]	0.3 [0.1, 0.9]	0.3 [0.1, 0.7]	2.5	-27.6 [-31.5, -24.8]	276
	processed (step 5)	-1.4 [-1.8, -0.5]	-1.4 [-1.9, -0.9]	-1.4 [-1.8, -0.7]	0.7	-33.3 [-36.6, -30.3]	276
60 Hz	**raw (step 1)**	4.8 [2.7, 9.9]	7.7 [2.5, 17.0]	5.5 [2.6, 12.9]	71.0	-23.1 [-27.6, -17.3]	276
	processed (step 5)	4.4 [2.7, 8.9]	6.6 [2.5, 15.1]	5.0 [2.6, 11.0]	69.2	-23.1 [-27.2, -18.1]	276
75 Hz	**raw (step 1)**	0.3 [0.1, 0.6]	0.3 [0.1, 0.6]	0.3 [0.1, 0.6]	2.2	-28.8 [-33.1, -25.5]	276
	processed (step 5)	0.6 [0.3, 0.9]	0.6 [0.3, 1.1]	0.6 [0.3, 1.0]	3.6	-28.5 [-33.1, -24.8]	276

Reading Figure 2.3, Figure 2.4 and Table 2.2. The **60 Hz** mains line is present in most raw recordings and is **not** removed by the pipeline (it is the same in the raw and processed rows of Table 2.2), because the notch of step 3 sits at 50 Hz (section 5). At **50 Hz** there is no line in the raw data; after the notch the processed spectra show a narrow dip (Figure 2.3(d)). Δ becomes only slightly negative, because the notch (±0.125 Hz) is narrower than the ±0.5 Hz search band of Eq. (2.5). At **75 Hz** there is no line either; it is only the start of the band-pass roll-off (Figure 4.1(c)), visible as the drop of the processed grand average in Figure 2.3(c, d).

3. Step 1 — channel selection & ordering

`make_TUAB.py` defines a list $\mathcal{D}$ of channels to drop (`drop_channels` : EKG, EMG, ROC/LOC, PHOTIC, IBI/BURSTS/SUPPR, RESP, `EEG 20-REF … EEG 128-REF` , …) and a fixed target order $\mathcal{O} = (\mathcal{O}_1, \ldots, \mathcal{O}_{23})$ of 23 channels (`chOrder_standard`):

$$\begin{aligned} \mathcal{C}_{\text{keep}} &= \mathcal{C}_{\text{raw}} \setminus \mathcal{D}, \qquad |\mathcal{C}_{\text{keep}}| = 23, \\ X^{(1)}_{i,:} &= X^{(0)}_{\pi(i),:}, \qquad \pi(i) = \text{index of } \mathcal{O}_i \text{ in } \mathcal{C}_{\text{raw}} \end{aligned} \tag{3.1}$$

where $\mathcal{C}_{\text{raw}}$ is the ordered channel list of the EDF and $X^{(0)}, X^{(1)}$ the signals before and after this step. If after dropping the file does not contain exactly these 23 channels, the recording is

skipped and its name written to `tuab-process-error-files.txt` (Table 11.2 counts them). Table 3.1 shows what is dropped for the two examples.

The 23 channels (Figure 3.1) are the 19 electrodes of the international 10-20 system (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz), the two ear electrodes **A1/A2** and the anterior temporal electrodes **T1/T2**. Later, LaBraM maps each name to its index in the 62-channel `standard_1020` list for the spatial position embedding (Table 8.1).

Table 3.1. Channel selection for the two examples

class	recording	raw channels	dropped	kept	dropped channels
normal	aaaaaayx_s002_t000	36	13	23	ROC, LOC, EKG1, EMG, 26, 27, 28, 29, 30, PHOTIC, IBI, BURSTS, SUPPR
abnormal	aaaaabdo_s003_t000	30	7	23	ROC, LOC, EKG1, PHOTIC, IBI, BURSTS, SUPPR

```
target order (23): FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, A
1, A2, FZ, CZ, PZ, T1, T2
```

Figure 3.1. The 23 TUAB channels kept for LaBraM

Schematic scalp positions (nose up) of the 23 channels kept by Eq. (3.1): 19 standard 10-20 electrodes and the 4 extra ones (ear A1/A2, anterior temporal T1/T2, drawn at the rim).

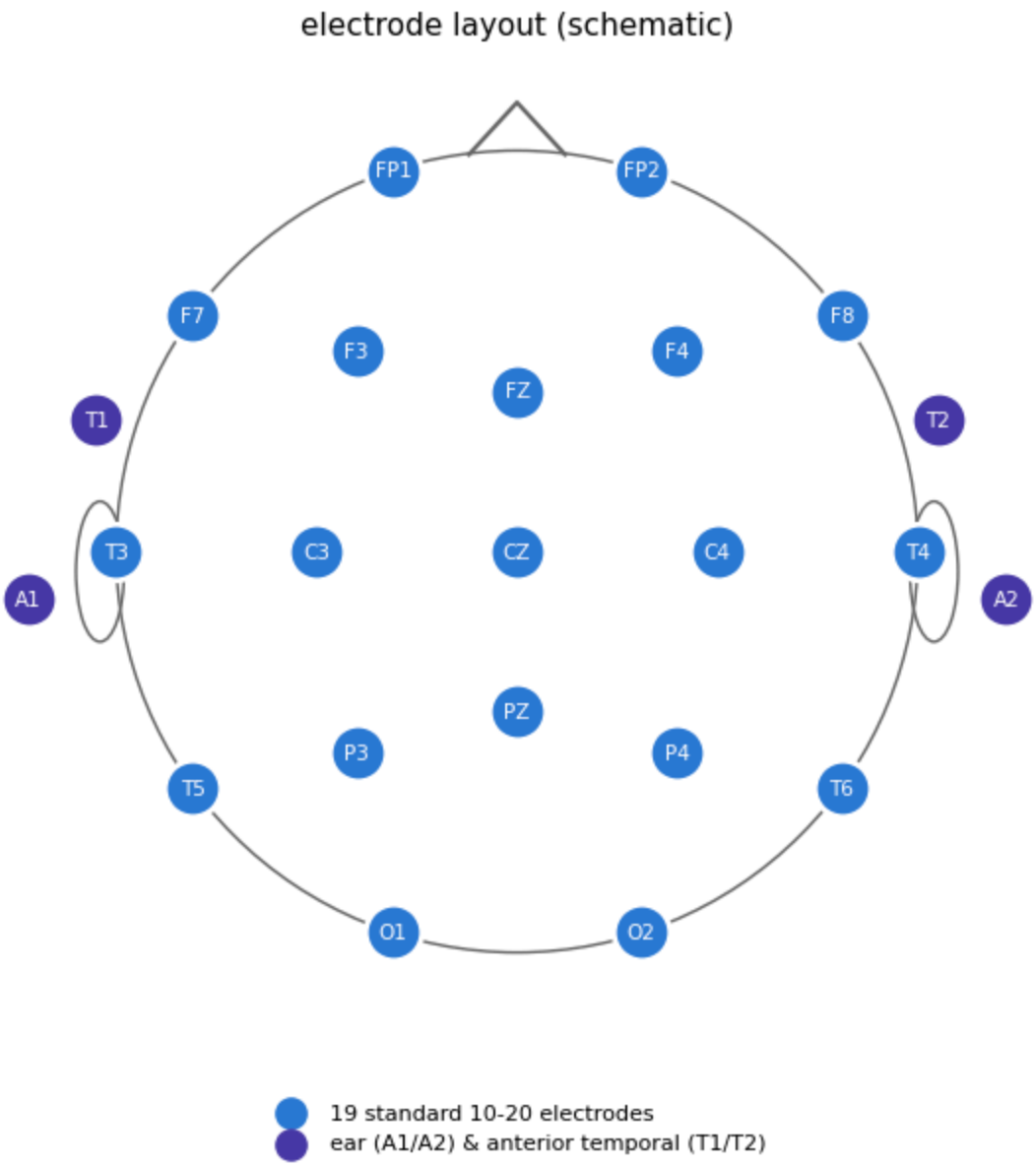
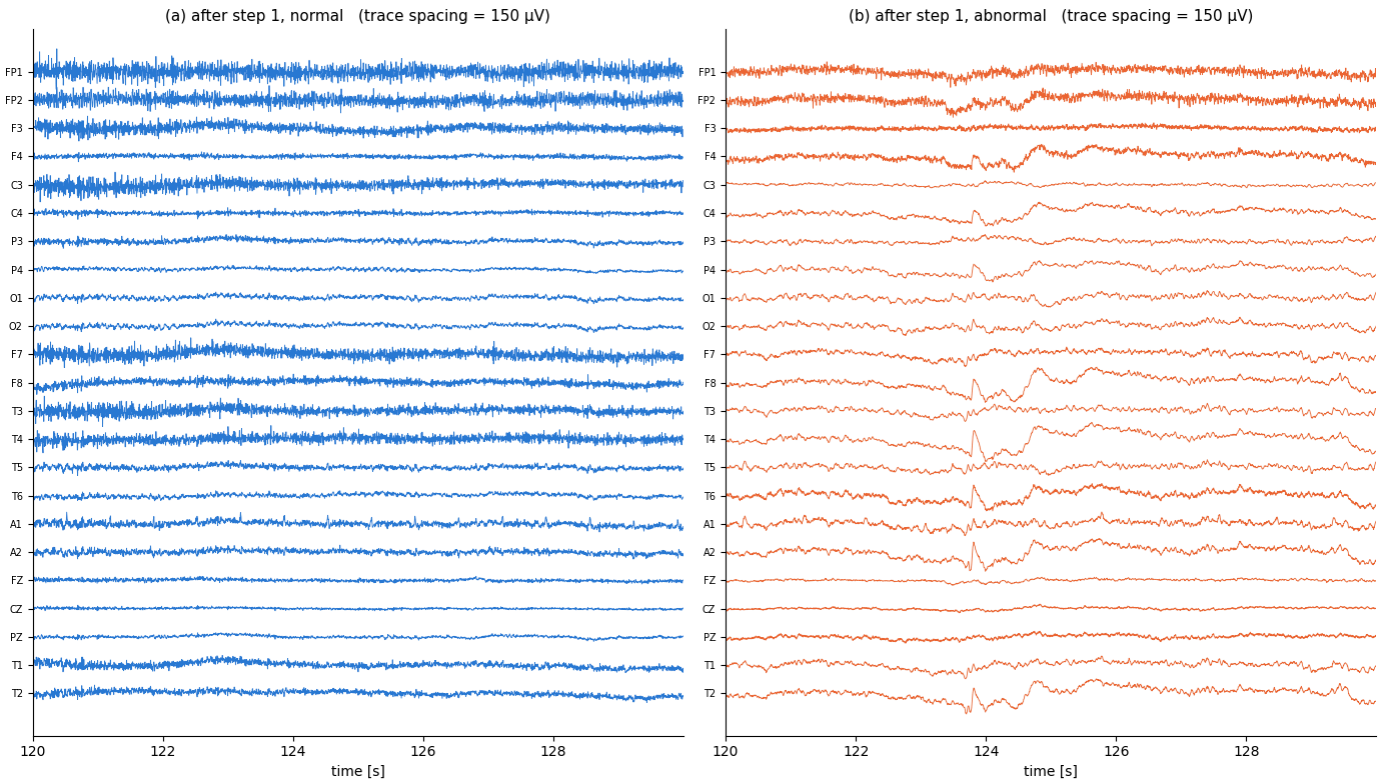


Figure 3.2. The 10 s window after channel selection (step 1)

The signal after step 1 — 23 EEG channels in the fixed order of Eq. (3.1), still 250 Hz and unfiltered — for (a) the normal and (b) the abnormal example (same 10 s as Figure 2.1, μV).



4. Step 2 — band-pass filter 0.1–75 Hz

`raw.filter(l_freq=0.1, h_freq=75.0)` uses MNE's defaults: a **linear-phase FIR** filter designed with the **window method** (`firwin` , Hamming window) and applied with **zero phase** (delay-compensated, non-causal).

Design. An ideal band-pass between f_1 and f_2 has the impulse response (difference of two ideal low-passes)

$$h_d[n] = \frac{2f_2}{f_s}\operatorname{sinc}\Big(\frac{2f_2 n}{f_s}\Big) - \frac{2f_1}{f_s}\operatorname{sinc}\Big(\frac{2f_1 n}{f_s}\Big), \qquad \operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u} \qquad (4.1)$$

where $f_s = 250$ Hz and f_1, f_2 are the −6 dB cut-offs (Hz). It is truncated to N taps and multiplied by a Hamming window:

$$h[n] = h_d\Big[n - \tfrac{N-1}{2}\Big]\, w[n], \qquad w[n] = 0.54 - 0.46\cos\Big(\frac{2\pi n}{N-1}\Big), \qquad n = 0, \ldots, N -$$

The cut-offs sit in the middle of the transition bands. MNE's automatic transition bandwidths for the pass-band edges $f_l = 0.1$ Hz and $f_h = 75$ Hz are

$$\begin{aligned} \Delta f_l &= \min\big(\max(0.25\, f_l,\ 2),\ f_l\big), & \Delta f_h &= \min\big(\max(0.25\, f_h,\ 2),\ \tfrac{f_s}{2} - f_h\big), \\ f_1 &= f_l - \tfrac{\Delta f_l}{2}, & f_2 &= f_h + \tfrac{\Delta f_h}{2} \end{aligned} \qquad (4.3$$

and the filter length for a Hamming window is

$$N = \left\lceil 3.3\,\frac{f_s}{\min(\Delta f_l, \Delta f_h)} \right\rceil \ \text{(rounded up to odd)} \tag{4.4}$$

Table 4.1 evaluates Eqs. (4.3)–(4.4). The very long filter (~33 s) is the price of the very narrow transition at the low end.

Application (zero phase). Since h is symmetric, centring it removes the $(N-1)/2$-sample delay:

$$y[n] = \sum_{k=0}^{N-1} h[k]\; x\!\left[n + \tfrac{N-1}{2} - k\right] \tag{4.5}$$

where x is the input (µV) and y the filtered output (µV); the signal is padded at both ends by reflection (`pad="reflect_limited"`) to reduce edge effects.

Why. Below 0.1 Hz there is no useful brain signal in a clinical EEG — only electrode offset, sweat and movement drift. Above ~75 Hz the scalp EEG is dominated by muscle (EMG) and instrument noise; 75 Hz also keeps the signal below the 100 Hz Nyquist frequency of the 200 Hz target rate (step 4).

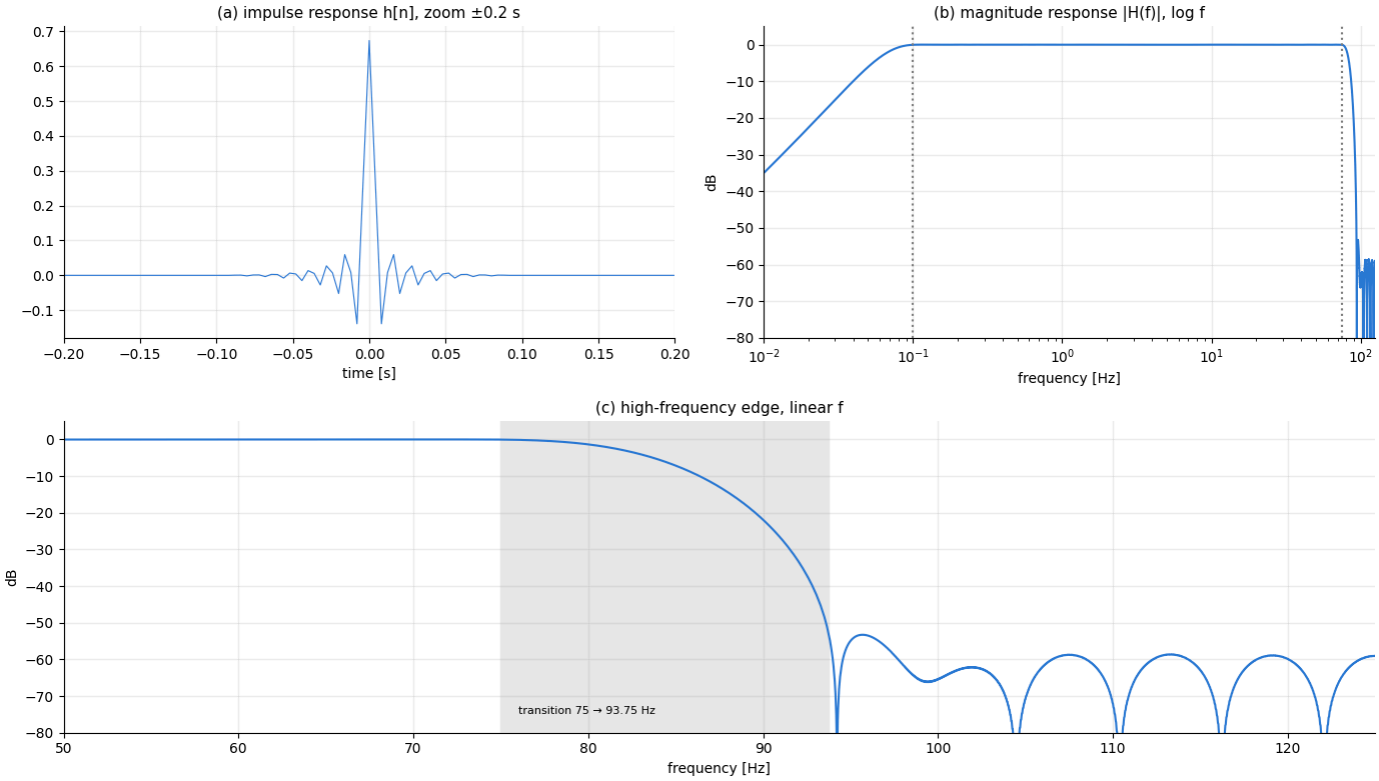
Table 4.1. Parameters of the band-pass filter

Computed from Eqs. (4.3)–(4.4) and checked against the filter MNE actually designs (`mne.filter.create_filter` , identical to `raw.filter`).

quantity	value	Eq.
pass-band edges f_l / f_h	0.1 / 75.0 Hz	(4.3)
transition bandwidths Δf_l / Δf_h	0.1 / 18.75 Hz	(4.3)
−6 dB cut-offs f_1 / f_2	0.05 / 84.375 Hz	(4.3)
stop band starts below / above	0 / 93.75 Hz	(4.3)
taps N (Eq.) / taps (MNE)	8251 / 8251	(4.4)
filter length	33.0 s	(4.4)
window / phase	Hamming / zero phase	(4.2), (4.5)

Figure 4.1. Band-pass FIR 0.1–75 Hz (Hamming, zero phase)

The band-pass filter of Eqs. (4.1)–(4.5) at $f_s = 250$ Hz: (a) impulse response $h[n]$ (zoom ±0.2 s), (b) magnitude response $20\log_{10}|H(f)|$ on a log-frequency axis (dotted: 0.1 and 75 Hz), (c) the high-frequency edge on a linear axis with the transition band shaded.



4.1 Before vs after the band-pass on the same 10 s of real EEG

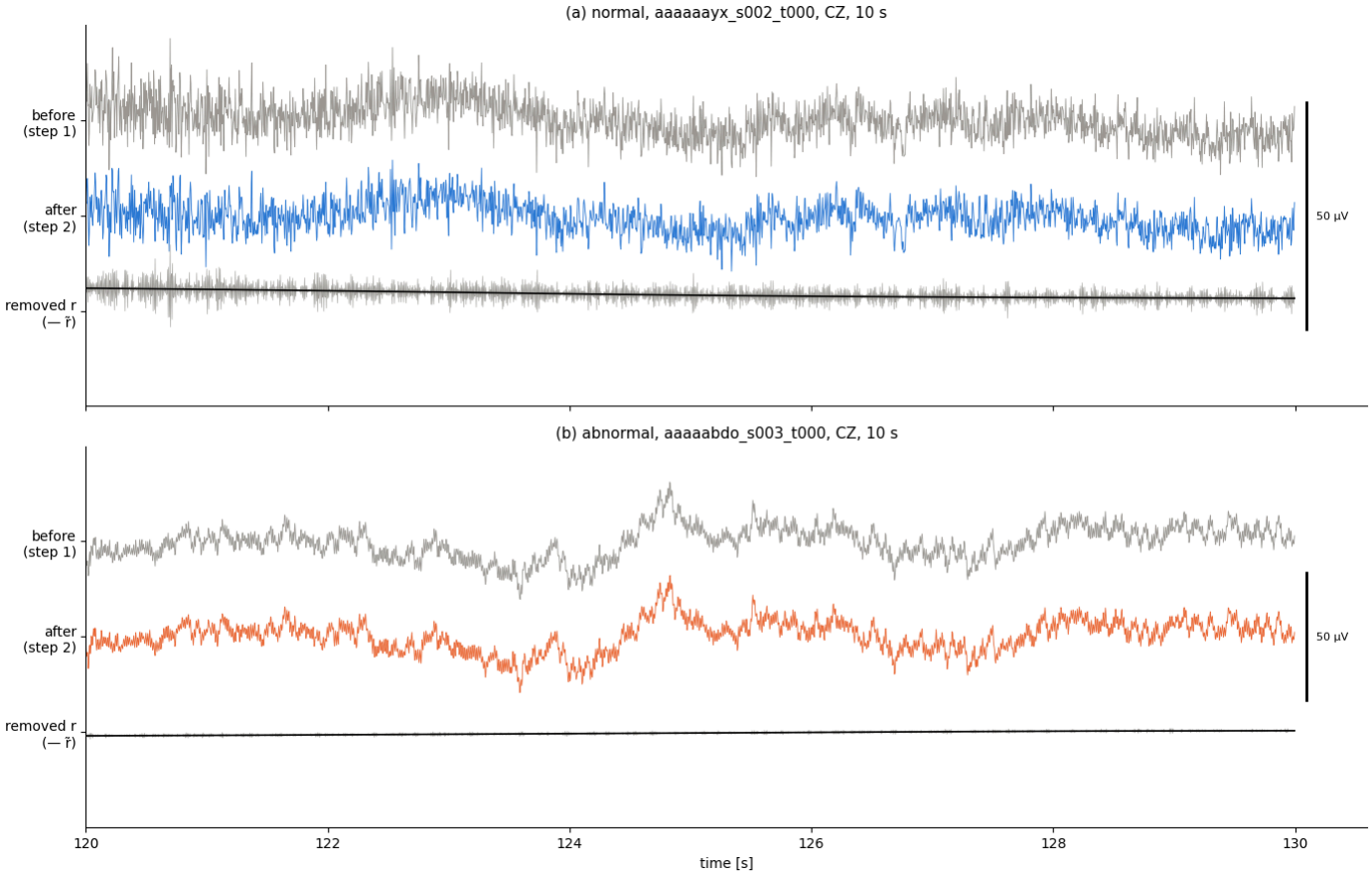
Figures 4.2–4.4 show the same **10 s window** as every other step (`SHOW_T0` … `SHOW_T0 + 10` s) before (step 1) and after (step 2) the band-pass, at the same vertical scale. What the filter removed is

$$r[n] = x[n] - y[n], \qquad \tilde{r}[n] = \frac{1}{f_s} \sum_{k=-f_s/2}^{f_s/2-1} r[n+k] \tag{4.6}$$

where x is the step-1 signal, y the band-passed signal of Eq. (4.5) (both µV), r the removed component and $\tilde{r}$ its 1 s moving average. Because the pass band is 0.1–75 Hz, $\tilde{r}$ is the **slow drift / DC offset** (< 0.1 Hz) and $r - \tilde{r}$ is mostly the **> 75 Hz** content (muscle activity, noise).

Figure 4.2. Cz before / after the band-pass and what was removed, 10 s

One channel (Cz), the 10 s window: before (grey, top), after (colour, middle) and the removed component r of Eq. (4.6) (bottom, with $\tilde{r}$ in black), for (a) the normal and (b) the abnormal example. Traces are offset vertically; the scale bar is 50 µV. No display de-meaning except the offsets, so drift is visible as wander of the top trace.



Figures 4.3 (normal) and 4.4 (abnormal) show all 23 channels of the same 10 s window, (a) before and (b) after the band-pass. Same trace spacing (150 µV) in both panels and in every per-step figure; per-channel mean of the window removed for display. The high-frequency fuzz (muscle) and the slow wander are gone in (b); the rhythmic brain activity is unchanged.

Figure 4.3. Normal example: 10 s before vs after the band-pass

`aaaaaayx_s002_t000`, all 23 channels: (a) before the band-pass (step 1), (b) after it (step 2); spacing 150 µV.

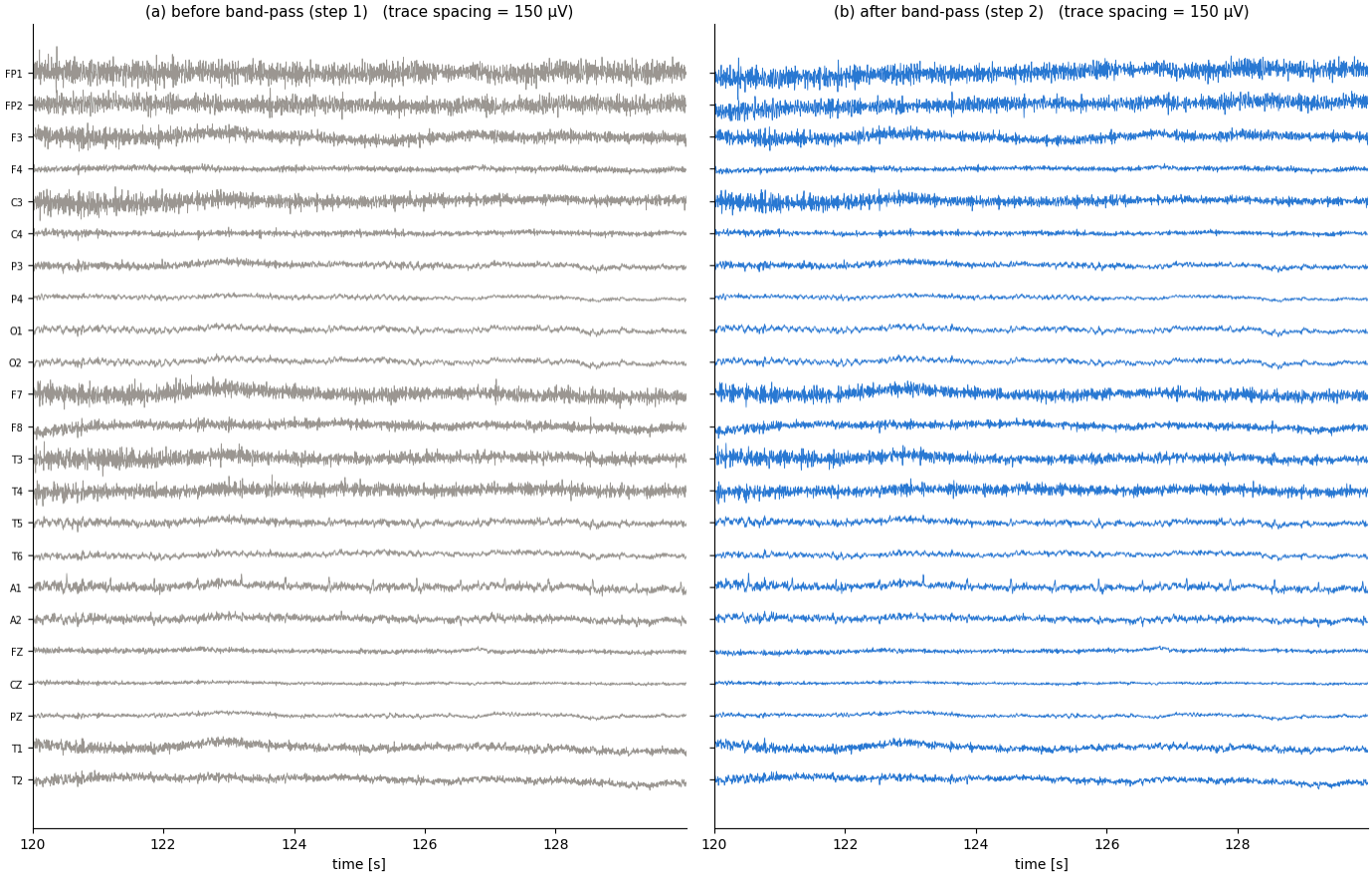


Figure 4.4. Abnormal example: 10 s before vs after the band-pass

`aaaaabdo_s003_t000`, all 23 channels: (a) before the band-pass (step 1), (b) after it (step 2); spacing 150 µV.

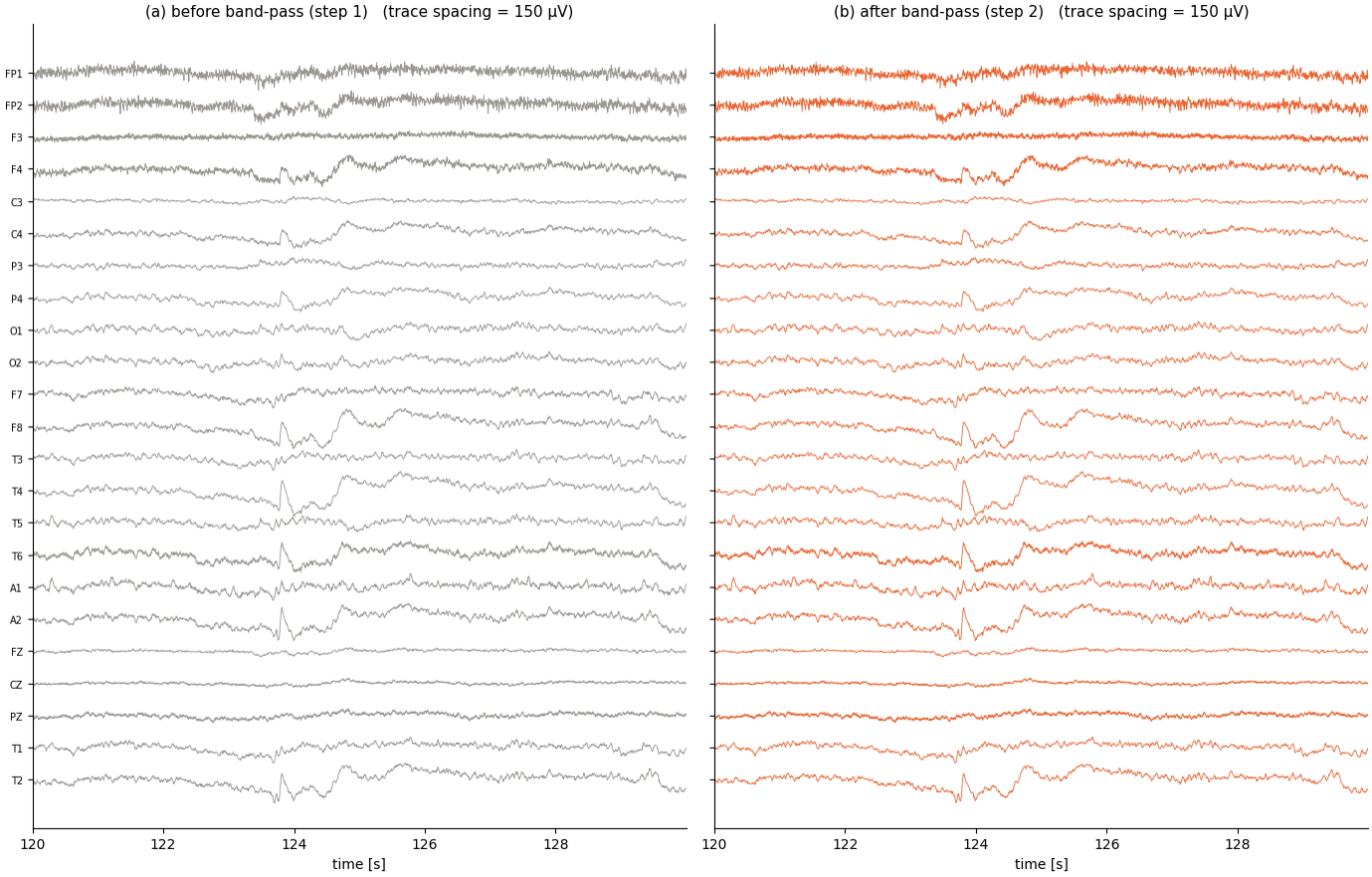


Figure 4.5. Removed component r over the whole recording

The removed component r of Eq. (4.6) for Cz over the **whole recording** (grey) and its 1 s moving average $\tilde{r}$ (colour) for (a) the normal and (b) the abnormal example (µV). Bursts of grey are muscle / movement artefacts (> 75 Hz content); the coloured line is the slow drift.

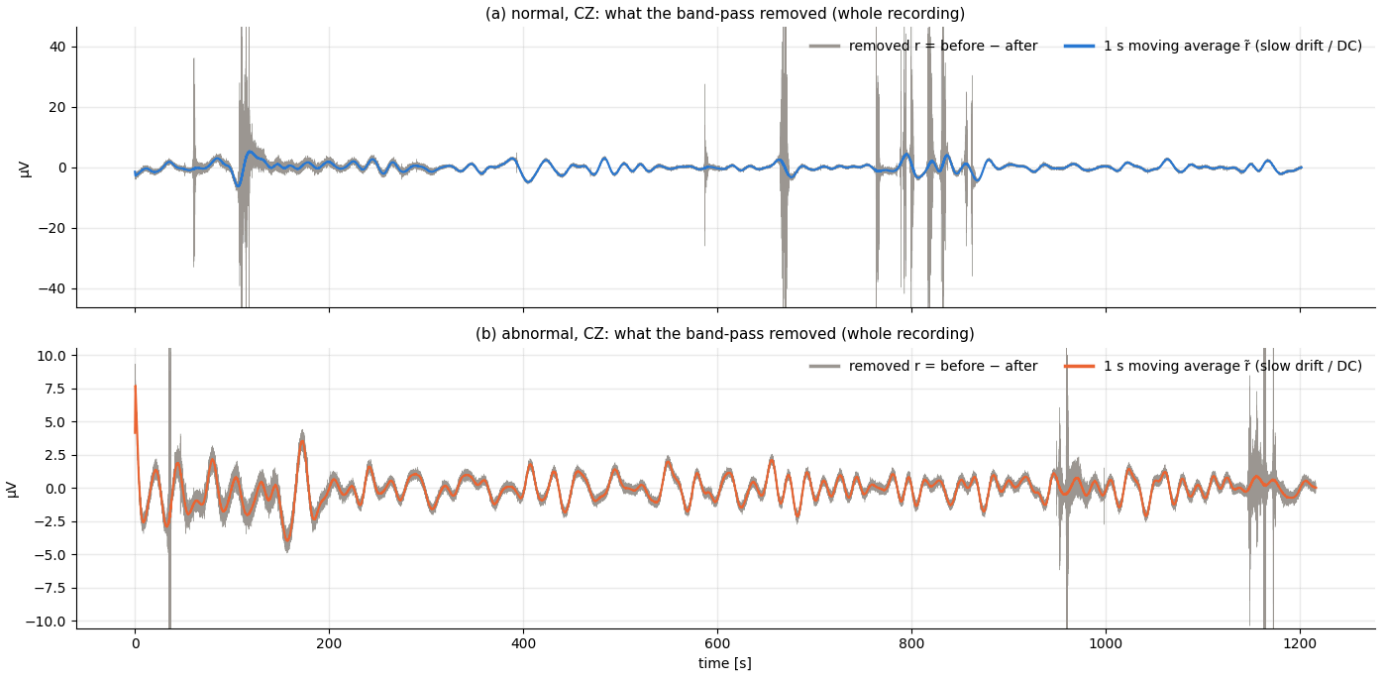
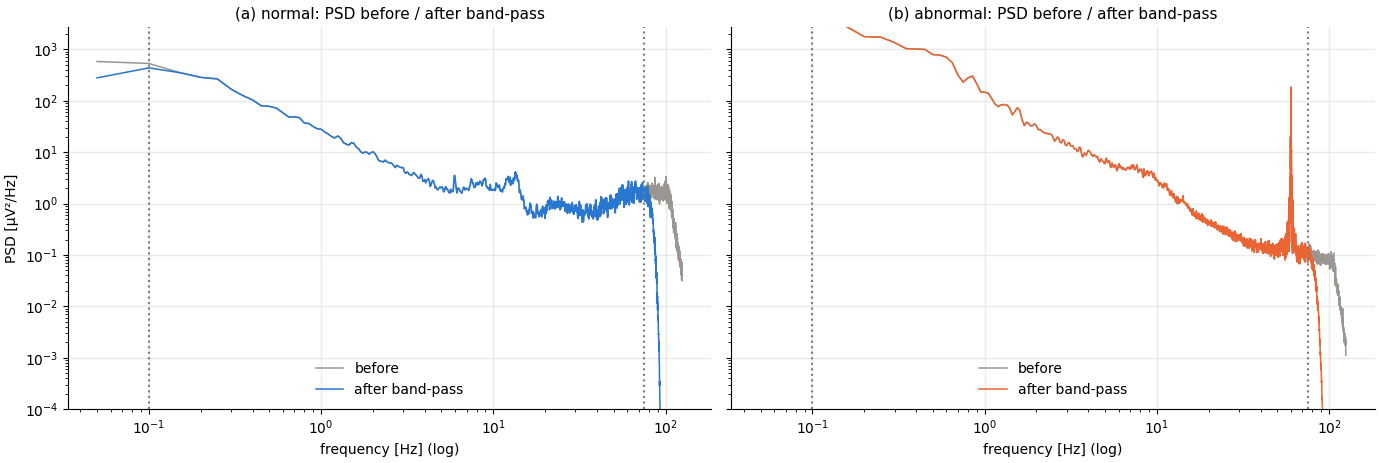


Figure 4.6. Spectrum before vs after the band-pass

Channel-averaged PSD, Eq. (2.3), before (grey) and after (colour) the band-pass for (a) the normal and (b) the abnormal example (whole recording, 20 s Welch segments, log–log axes; dotted: 0.1 and 75 Hz).



5. Step 3 — 50 Hz notch filter

`raw.notch_filter(50.0)` builds a narrow FIR **band-stop** filter (same window method, Hamming, zero phase as Eqs. (4.2), (4.5)). With the defaults `notch_widths` $w = f_0/200 = 0.25$ Hz and `trans_bandwidth` $\Delta f = 1$ Hz:

$$h_{\text{notch}}[n] = \delta[n] - h_{\text{bp}}^{[f_0 - w/2,\, f_0 + w/2]}[n], \qquad |H_{\text{notch}}(f)| \approx \begin{cases} 0, & |f - f_0| \le w/2 \\ 1, & |f - f_0| \ge w/2 + \Delta f/ \end{cases}$$

where $f_0 = 50$ Hz, δ the unit impulse and $h_{\text{bp}}^{[a,b]}$ a band-pass of Eq. (4.2) between a and b. It removes a sliver of spectrum 50 ± 0.125 Hz and leaves everything outside ≈ 49.4–50.6 Hz untouched (Figure 5.1(a)).

> ⚠️ **TUH EEG was recorded in the USA, where the mains frequency is 60 Hz, not 50 Hz.** The 50 Hz notch therefore does essentially nothing on TUAB, and because the band-pass keeps everything up to 75 Hz, the **60 Hz line noise stays in the training data** (Figure 5.1(b), Table 5.1, and for the whole test set Table 2.2). The script was written for a mix of datasets; it is kept as-is here because the published checkpoints were trained on data processed exactly this way — changing it would change the input distribution.

Figure 5.1. The 50 Hz notch and the surviving 60 Hz line

(a) Magnitude response of the 50 Hz notch, Eq. (5.1), at 250 Hz (dB); (b) channel-averaged PSD, Eq. (2.3), around the line frequencies before (dotted) and after (solid) the notch for both

examples.

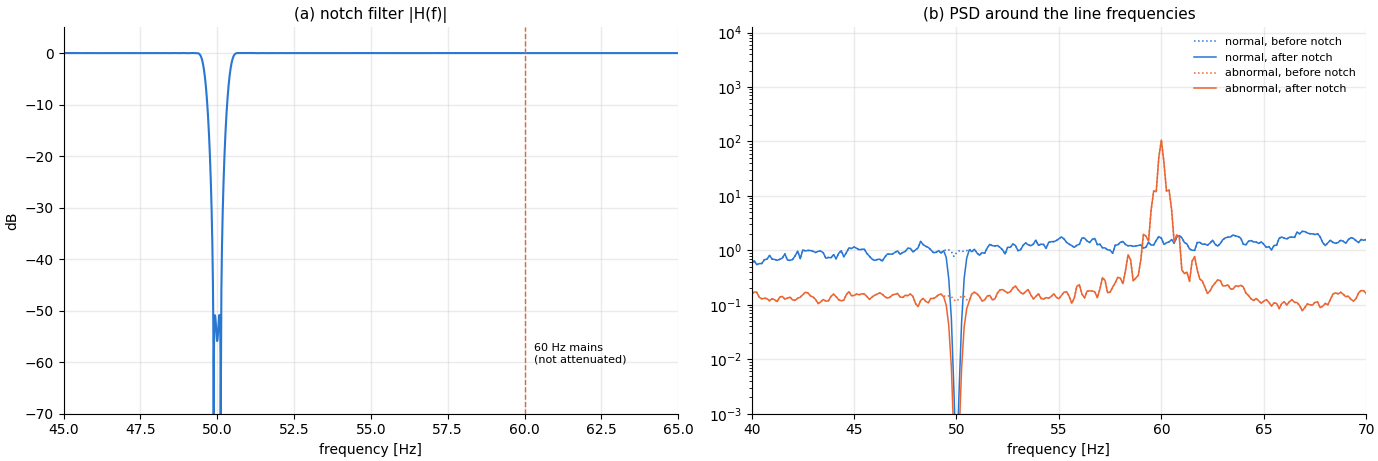


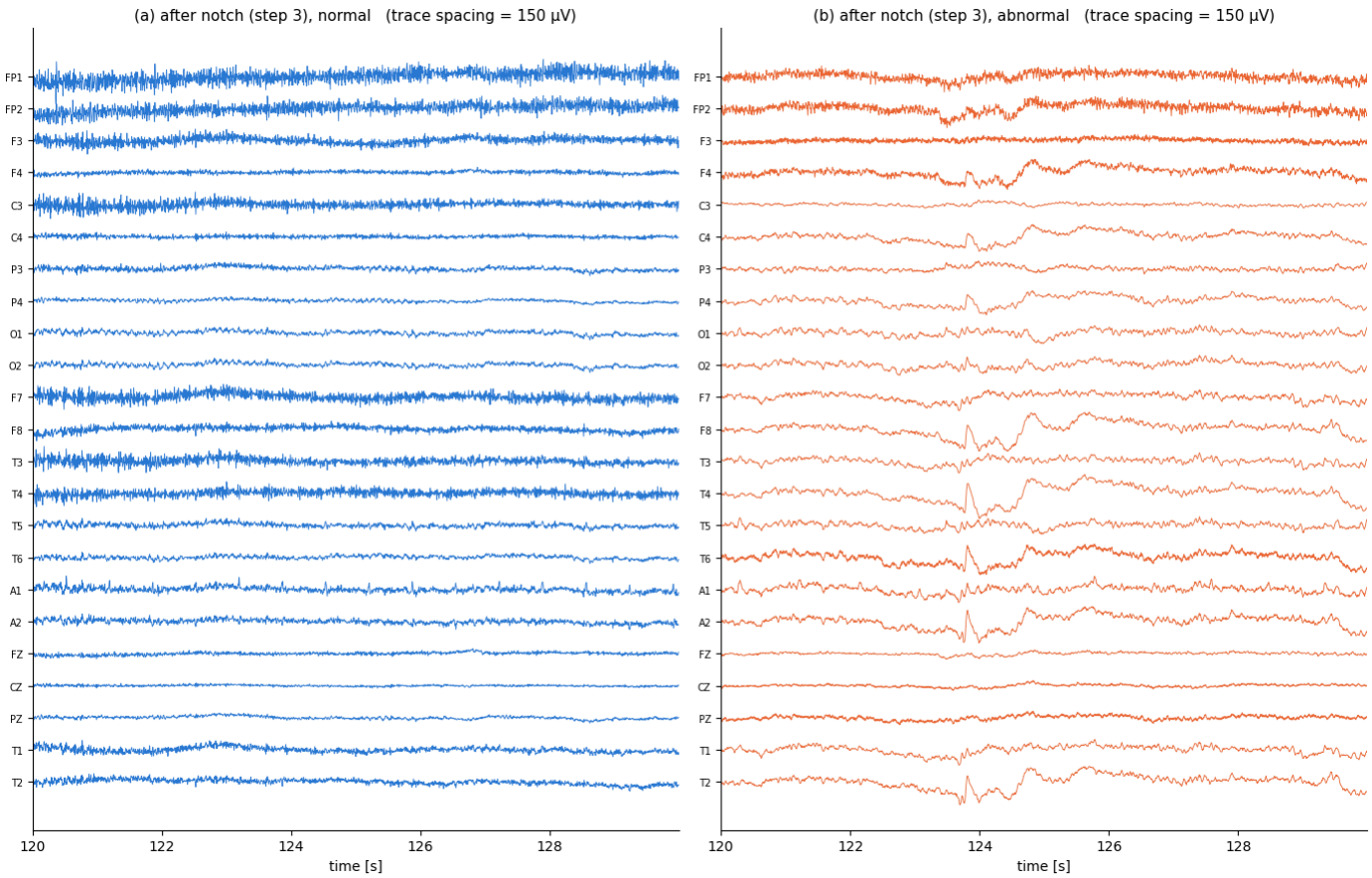
Table 5.1. Line prominence before and after the notch for the two examples

Prominence $\Delta(f_0)$ of Eq. (2.5) in dB (8 s Welch segments), before (step 2) and after (step 3) the notch.

class	recording	Δ(50 Hz) before [dB]	Δ(50 Hz) after [dB]	Δ(60 Hz) before [dB]	Δ(60 Hz) after [dB]
normal	aaaaaayx_s002_t000	-0.3	-1.7	1.3	1.3
abnormal	aaaaabdo_s003_t000	0.0	-1.7	26.9	26.9

Figure 5.2. The 10 s window after the 50 Hz notch (step 3)

The same 10 s window after step 3 (notch) for (a) the normal and (b) the abnormal example (23 channels, 250 Hz, µV, spacing 150 µV). Visually identical to Figures 4.3(b)/4.4(b): the notch removes only 50 ± 0.125 Hz, where TUAB has no line (Table 5.1).



6. Step 4 — resampling 250 → 200 Hz

LaBraM works with **200 samples per second** (one patch = 200 samples = 1 s) for all datasets. `raw.resample(200)` uses MNE's **FFT (Fourier) method** with a boxcar window: the signal of length N is transformed, its spectrum is truncated to the new Nyquist band, and transformed back with the new length N':

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}, \qquad N' = \operatorname{round}\!\left(N\,\frac{f_s'}{f_s}\right) = 0.8\,N \tag{6.1}$$

$$x'[m] = \frac{N'}{N}\cdot\frac{1}{N'}\sum_{|k|<N'/2} X[k]\, e^{\,j2\pi km/N'}, \qquad m = 0,\ldots,N'-1 \tag{6.2}$$

where x (µV, $f_s = 250$ Hz) is the notch-filtered signal and x' (µV, $f_s' = 200$ Hz) the resampled one; the factor N'/N keeps amplitudes unchanged. (The signal is first padded — `npad="auto"` to a fast FFT length, edges by reflection.) Truncating the spectrum at the new Nyquist frequency $f_s'/2 = 100$ Hz is an ideal anti-aliasing low-pass. Because step 2 already removed everything above ~85 Hz (≥ 53 dB attenuation from 93.75 Hz, Table 4.1), nothing is aliased and the spectrum below 75 Hz is unchanged (Figure 6.1(b, d)); Table 6.1 checks $N' = 0.8N$.

Figure 6.1. FFT resampling 250 → 200 Hz

Resampling for the normal (top) and abnormal (bottom) example: (a, c) 0.5 s of Cz with the 250 Hz samples (grey dots) and the 200 Hz samples (squares) — 5 samples ↔ 4 samples, same waveform; (b, d) channel-averaged PSD, Eq. (2.3), before and after (dotted: new Nyquist 100 Hz).

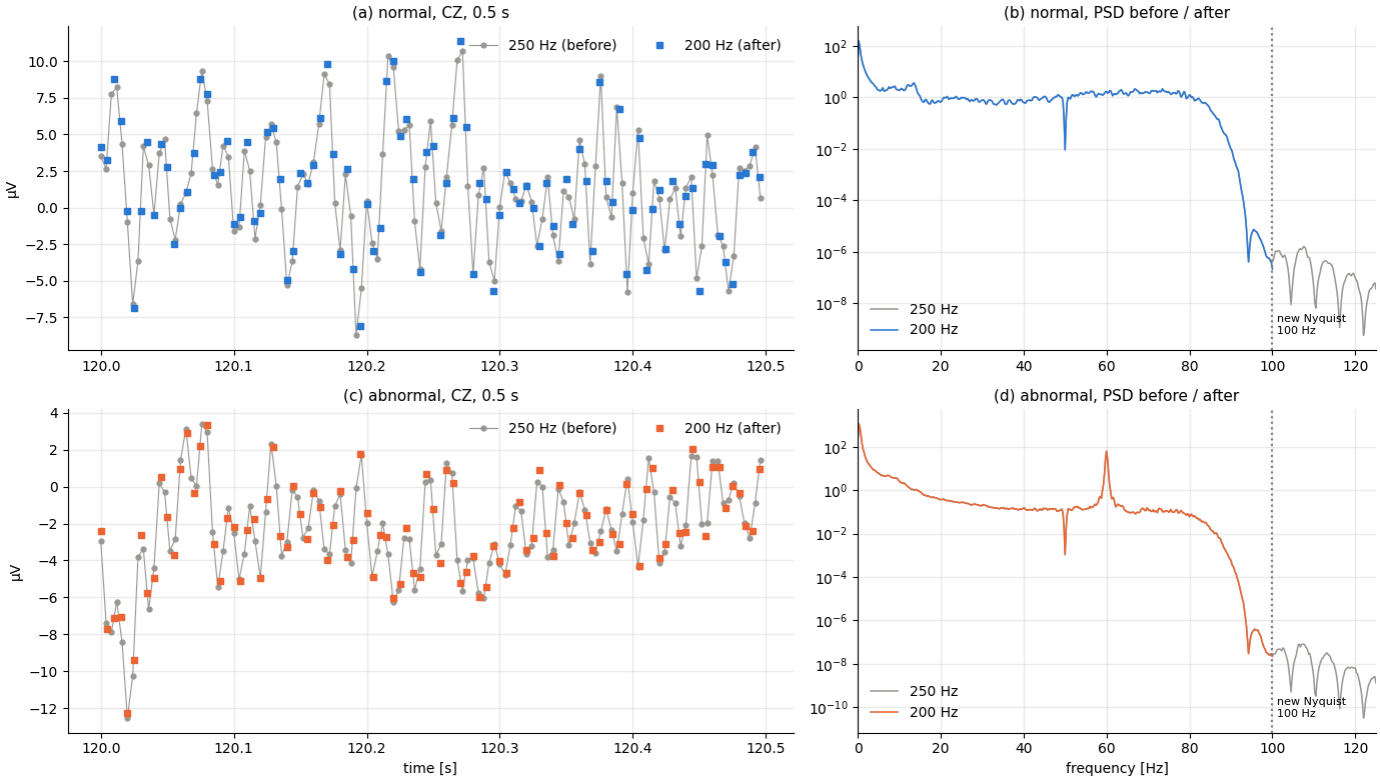


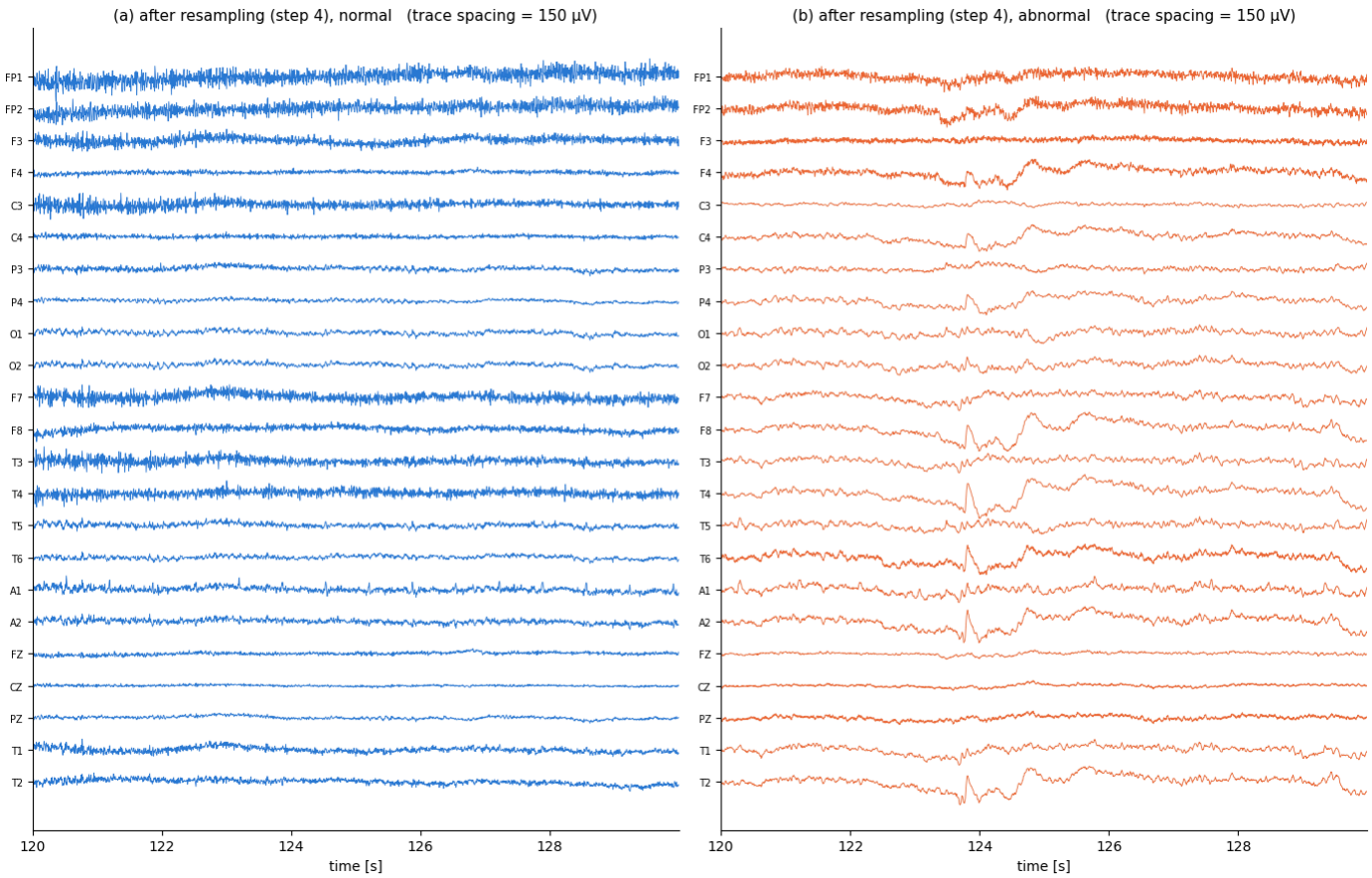
Table 6.1. Sample counts before and after resampling

N and N' of Eq. (6.1) for the two examples.

class	N (250 Hz)	N' (200 Hz)	N'/N	duration [s]
normal	300500	240400	0.8000	1202.0000
abnormal	304250	243400	0.8000	1217.0000

Figure 6.2. The 10 s window after resampling to 200 Hz (step 4)

The same 10 s window after step 4 (resampled to 200 Hz, 2000 samples per channel) for (a) the normal and (b) the abnormal example (µV, spacing 150 µV).



7. Step 5 — windowing into 10 s samples, labels, verification

After resampling, the recording $X \in \mathbb{R}^{23\times N'}$ (µV, 200 Hz) is cut into **non-overlapping** windows of $W = 2000$ samples (10 s):

$$K = \left\lfloor \frac{N'}{W} \right\rfloor, \qquad X_k = X_{:,\, kW:(k+1)W} \in \mathbb{R}^{23\times 2000}, \qquad y_k = y, \qquad k = 0, \ldots, K-1$$

where K is the number of windows, y the recording label of section 1, and the last $N' \bmod W$ samples (< 10 s) are discarded (Figure 7.1). Every window is saved as

```
pickle.dump({"X": X_k, "y": y}, open(f"processed/<split>/<stem>_<k>.pkl", "wb"))
```

There is **no** artefact rejection, re-referencing or amplitude normalisation in this offline step — windows containing movement or electrode artefacts (the large bursts in Figure 7.1(a), clipped by the y-axis) are kept as they are.

Figure 7.1. Windowing a whole recording into 10 s samples

Whole recording, Cz after step 4 (µV), with the 10 s window boundaries of Eq. (7.1) (grey lines) and the discarded tail (shaded), for (a) the normal and (b) the abnormal example.

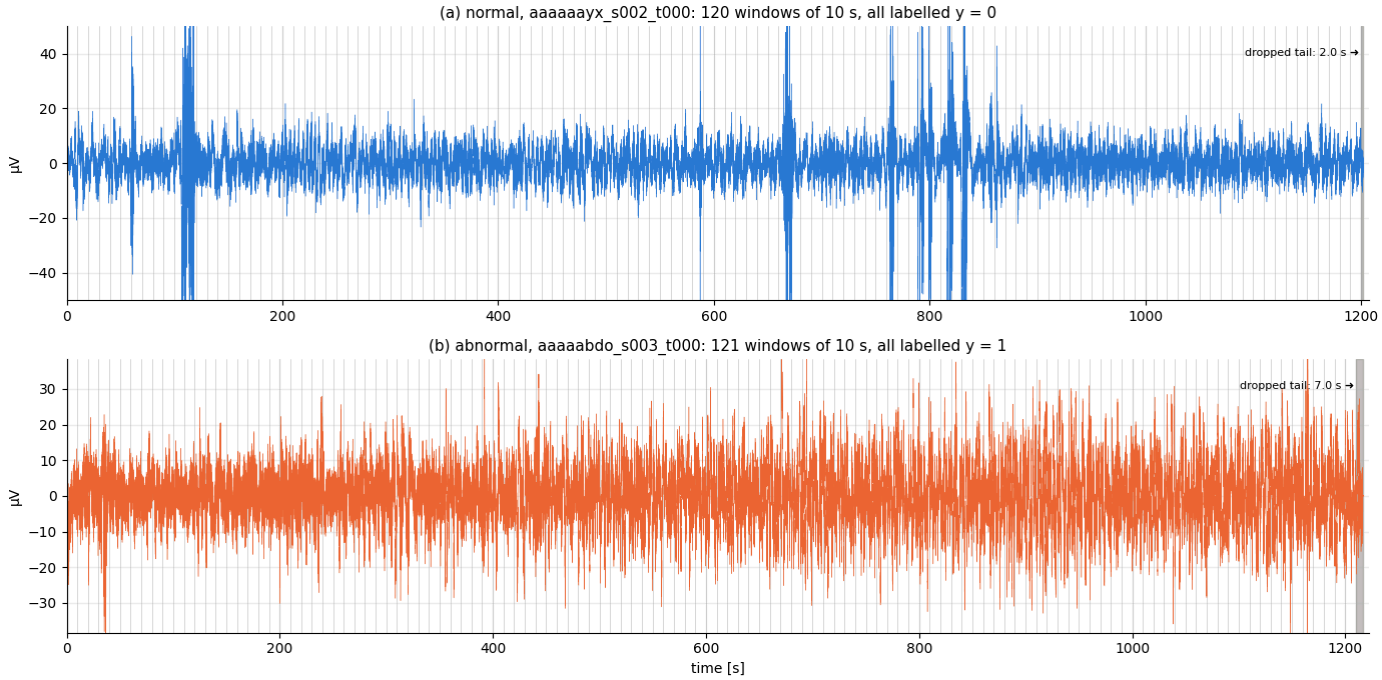


Table 7.1. Notebook re-implementation vs the pickles stored by `make_TUAB.py`

`make_TUAB.py` puts each recording in exactly one of `processed/{train,val,test}`. For every window k of the two examples we compare the stored array X_k^{stored} with ours, X_k of Eq. (7.1):

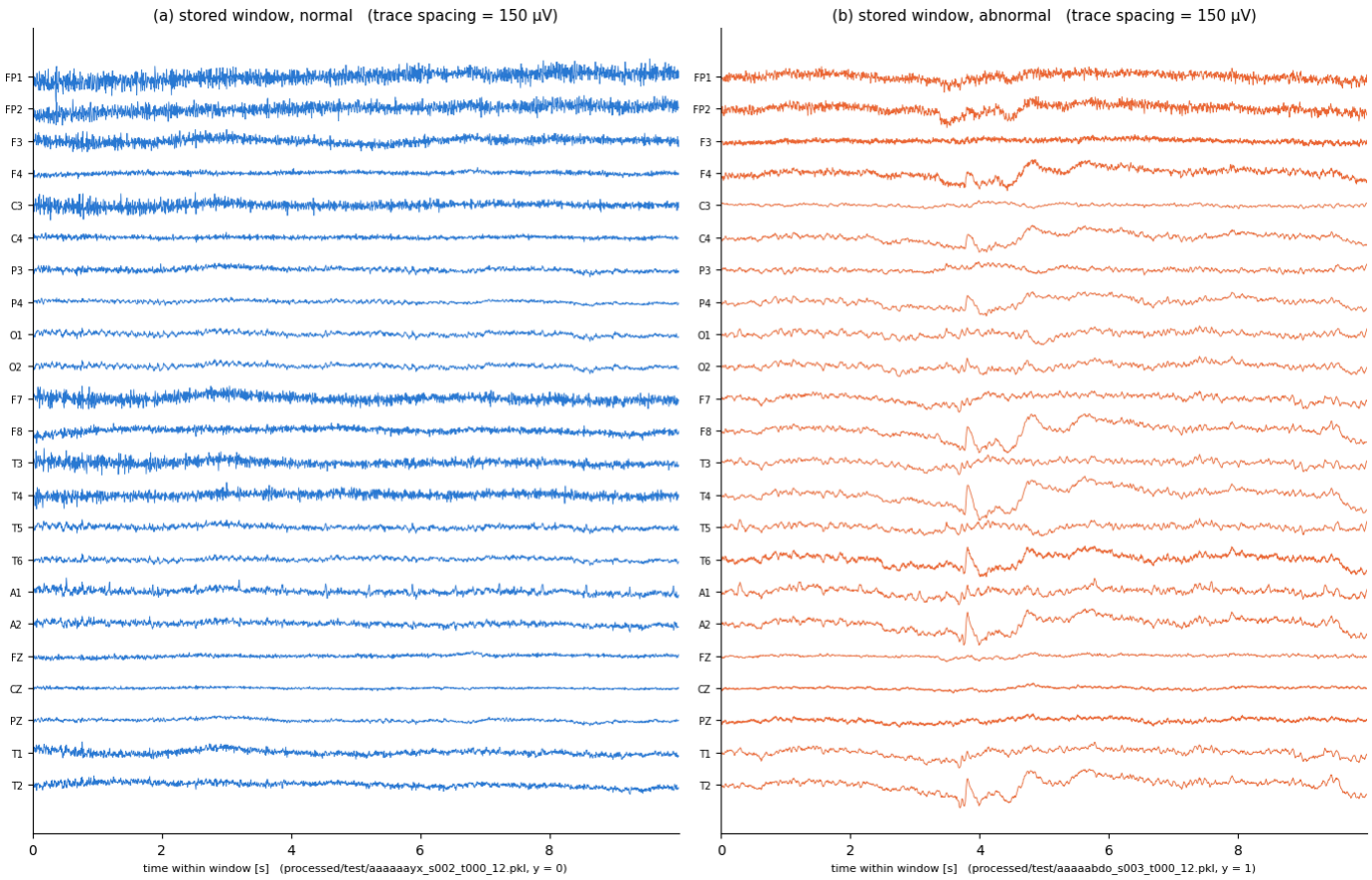
$$\varepsilon = \max_{k} \; \max_{c,t} \; \left| X^{\text{stored}}_{k,c,t} - X_{k,c,t} \right| \tag{7.2}$$

in µV; `stored y` lists the labels found in the pickles and `extra window on disk` flags a window beyond K.

class	recording	split	windows K	ε [µV], Eq. (7.2)	stored y	extra window on disk
normal	aaaaaayx_s002_t000	test	120	3.4e-13	[0]	False
abnormal	aaaaabdo_s003_t000	test	121	1.8e-12	[1]	False

Figure 7.2. The stored 10 s window (step 5)

The stored pickle of the same 10 s window, `<stem>_<k>.pkl` with k = `SHOW_T0`/10, read back from `processed/` (step 5, the array the loader returns), for (a) the normal and (b) the abnormal example (µV, spacing 150 µV). It equals Figure 6.2 to within $arepsilon$ of Table 7.1.



8. Step 6 — what the model actually receives

At training/evaluation time the `TUABLoader` reads a pickle and returns `(torch.FloatTensor(X), y)` (it would resample to another rate only if `sampling_rate ≠ 200`). Then `train_finetune.py` does two more things:

```
samples = samples.float() / 100                                  # µV ->
units of 100 µV
samples = rearrange(samples, 'B N (A T) -> B N A T', T=200)      # 1 s
patches
```

$$\tilde{x}_{c,a,t} = \frac{X_{c,\ 200a+t}}{100}, \qquad c = 1, \ldots, 23, \quad a = 0, \ldots, 9, \quad t = 0, \ldots, 199 \qquad (8.1)$$

where X is one stored window (µV), a the patch index and t the sample within the patch; $\tilde{x}$ is in units of 100 µV.

- **Scaling by 1/100** brings typical EEG amplitudes (tens of µV) to $O(0.1\text{–}1)$ — a fixed, global scale (no per-sample normalisation), identical to pre-training.
- **Patching**: each channel's 10 s window becomes 10 patches of 1 s → 23 × 10 = 230 tokens. Each patch is embedded by the temporal convolution; LaBraM adds a **temporal embedding** (patch index a) and a **spatial embedding** chosen by the electrode's position

in the 62-channel `standard_1020` list (`labram.data.eeg_constants.get_channel_indices`):

$$\iota_c = 1 + \text{index}\left(\text{name}_c \text{ in } \texttt{standard_1020}\right), \qquad \iota_0 = 0 \ (\text{CLS token}) \tag{8.2}$$

Table 8.1 lists ι_c. Figure 8.1(b) colours each token by its patch RMS,

$$\text{RMS}_{c,a} = \sqrt{\frac{1}{200}\sum_{t=0}^{199} \tilde{x}_{c,a,t}^{\,2}} \tag{8.3}$$

in units of 100 µV.

Table 8.1. Spatial-embedding index of each TUAB channel

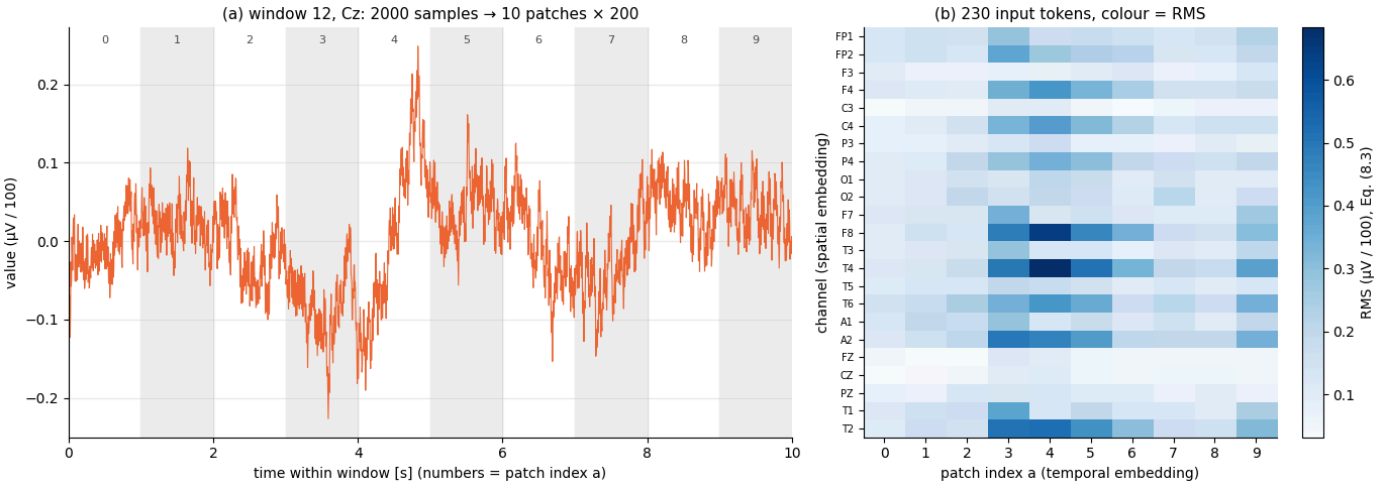
`ι` = Eq. (8.2); the CLS token has index 0.

channel	CLS	FP1	FP2	F3	F4	C3	C4	P3	P4	O1	O2	F7	F8	T3	T4	T5	T6	A1	A2	FZ	CZ	PZ	T1	T2
ι	0	1	3	18	22	40	44	62	66	81	83	16	24	89	91	90	92	95	96	20	42	64	113	114

Figure 8.1. Model input of the abnormal example's window (step 6)

Model input for one window of the abnormal example (the 10 s starting at `SHOW_T0`): (a) Cz after Eq. (8.1) (units of 100 µV), alternate 1 s patches shaded; (b) the 23 × 10 = 230 input tokens, coloured by $\text{RMS}_{c,a}$ of Eq. (8.3).

```
one window (23, 2000) µV  ->  model input (1, 23, 10, 200) (B, channels, patches, samples)
value range after /100: min -1.17, max 0.88, std 0.208
```



9. Step 7 (optional) — LaBraM++ preprocessing: CAR + per-patch z-score

Off by default. With `labram_plus.enabled=true` (see `docs/labram_plus_plus.md`) the model itself applies two more transforms to every patch tensor $x \in \mathbb{R}^{B\times N\times A\times T}$ (`labram/data/preprocess.py`), first CAR then z-scoring.

Common average reference (`common_average_reference`) — subtract the instantaneous mean over the N electrodes:

$$x^{\mathrm{CAR}}_{b,c,a,t} = x_{b,c,a,t} - \frac{1}{N}\sum_{c'=1}^{N} x_{b,c',a,t} \tag{9.1}$$

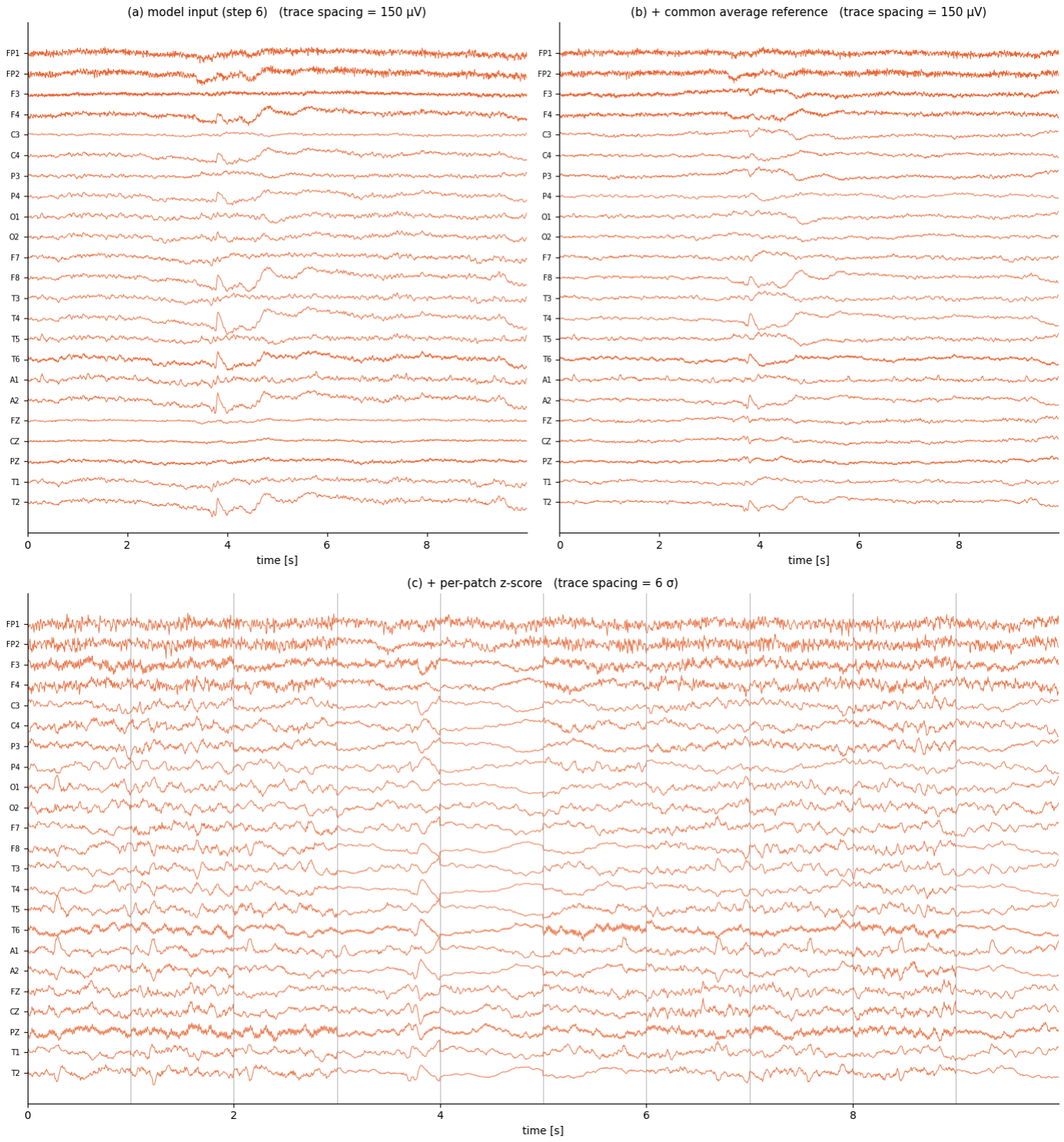
Per-patch z-score (`z_score_per_patch`) — standardise each (channel, patch) over its $T = 200$ samples:

$$\begin{aligned} z_{b,c,a,t} &= \frac{x^{\mathrm{CAR}}_{b,c,a,t} - \mu_{b,c,a}}{\sigma_{b,c,a} + \varepsilon}, \\ \mu_{b,c,a} &= \frac{1}{T}\sum_{t} x^{\mathrm{CAR}}_{b,c,a,t}, \qquad \sigma_{b,c,a} = \sqrt{\frac{1}{T-1}\sum_{t}\left(x^{\mathrm{CAR}}_{b,c,a,t} - \mu_{b,c,a}\right)^2} \end{aligned} \tag{9.2}$$

where b is the batch index, c the channel, a the patch, t the sample, and $\varepsilon = 10^{-5}$; z is unitless (σ units). Eq. (9.2) removes per-patch amplitude information entirely (every patch has unit variance) — it equalises channels but hides "this patch is larger than usual" (Figure 9.1(c)).

Figure 9.1. LaBraM++ preprocessing of the same window (step 7)

LaBraM++ preprocessing of the window of Figure 8.1: (a) model input of Eq. (8.1) (shown ×100, μV), (b) after CAR, Eq. (9.1) (μV), (c) after the per-patch z-score, Eq. (9.2) (σ; grey lines = patch boundaries).



10. Normal vs abnormal: examples and class-level spectra

10.1 Gallery

Three normal and three abnormal recordings from the test set (Figures 10.1–10.6). For each we show (a) the **raw** signal (step 0, 23 EEG channels only, 250 Hz) next to (b) the **final 10 s window** stored on disk (after step 5), at the same trace spacing.

Which window to show? The first window is biased toward set-up artefacts, and the cleanest one would be cherry-picking. We show a **typical window**:

$$k^* = \text{argmedian}_k \; q_{99}\big(|X_k|\big) \tag{10.1}$$

where $q_{99}\big(|X_k|\big)$ is the 99th percentile of the absolute amplitude (µV) of window k over all channels and samples, and argmedian picks the window whose value is the median over the recording.

What to look for (clinical heuristics, not a rule — many abnormal EEGs look normal in any given 10 s): normal adult EEG typically has a clear **posterior alpha rhythm (8–13 Hz)** at O1/O2/P3/P4 during relaxed wakefulness; abnormal recordings more often show **slowing** (excess delta 0.5–4 Hz / theta 4–8 Hz), focal or asymmetric activity, or epileptiform discharges (spikes / sharp waves). Raw and processed windows look **very similar** at this scale: the pipeline removes drift, > 75 Hz content and resamples — it does not "clean" the EEG of artefacts.

Figures 10.1–10.6 show one figure per recording (10.1–10.3 normal, 10.4–10.6 abnormal): (a) raw, step 0, at the time of the typical window k^* of Eq. (10.1); (b) the stored window `<stem>_<k*>.pkl` (µV).

Figure 10.1. Normal recording aaaaaayx_s002_t000

120 windows; typical window k* = 29 (Eq. (10.1)): (a) raw, (b) stored.

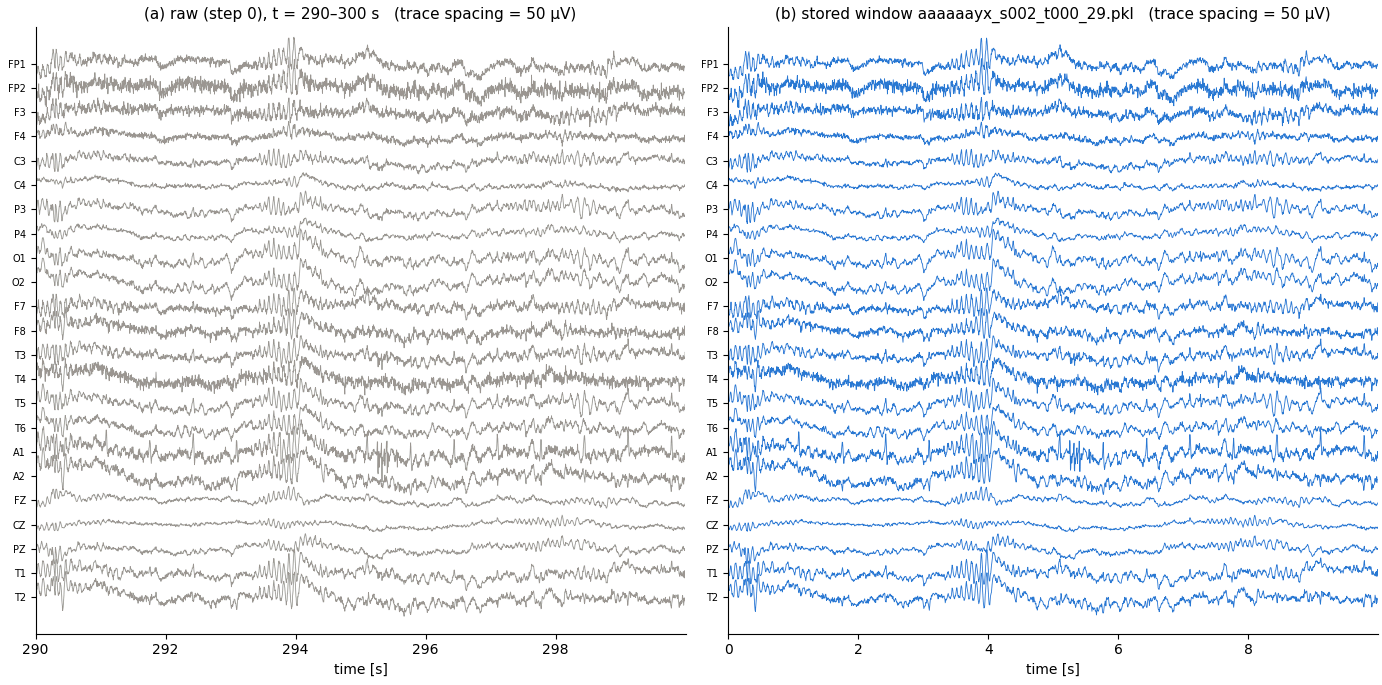


Figure 10.2. Normal recording aaaaacad_s003_t000

125 windows; typical window k* = 51 (Eq. (10.1)): (a) raw, (b) stored.

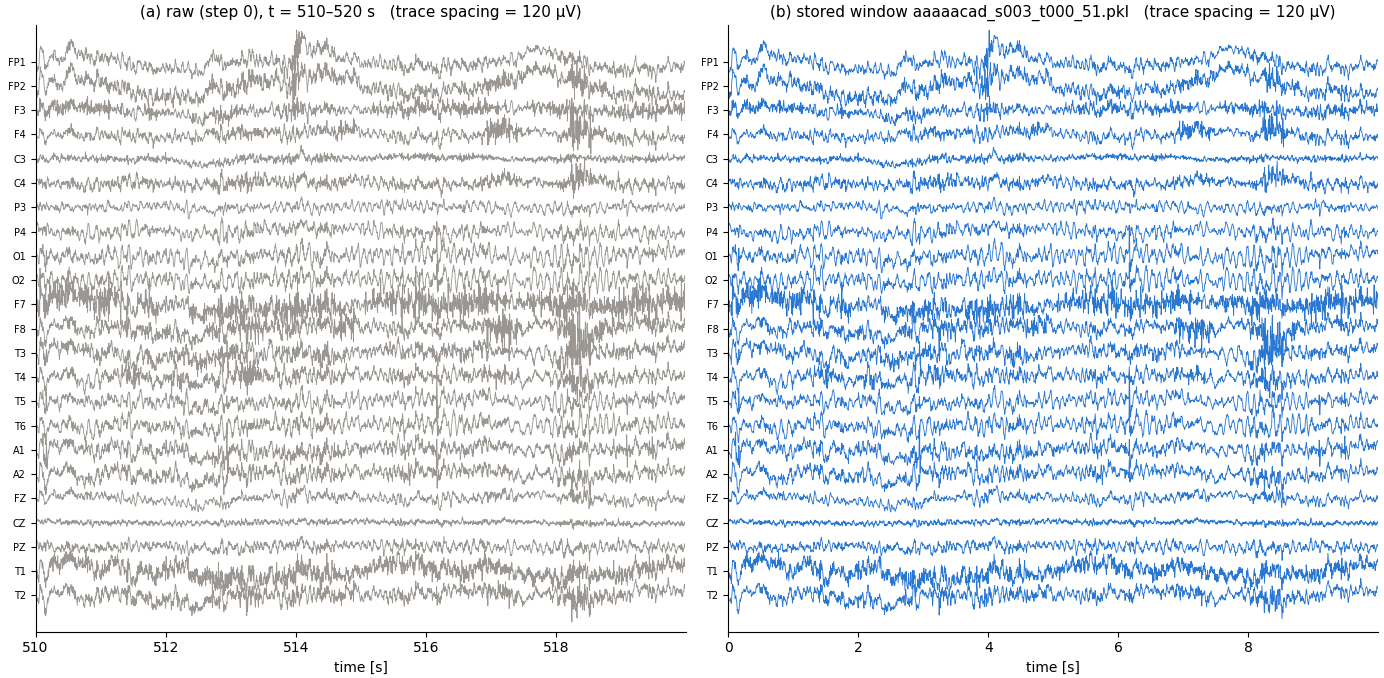


Figure 10.3. Normal recording aaaaacby_s004_t000

120 windows; typical window k* = 26 (Eq. (10.1)): (a) raw, (b) stored.

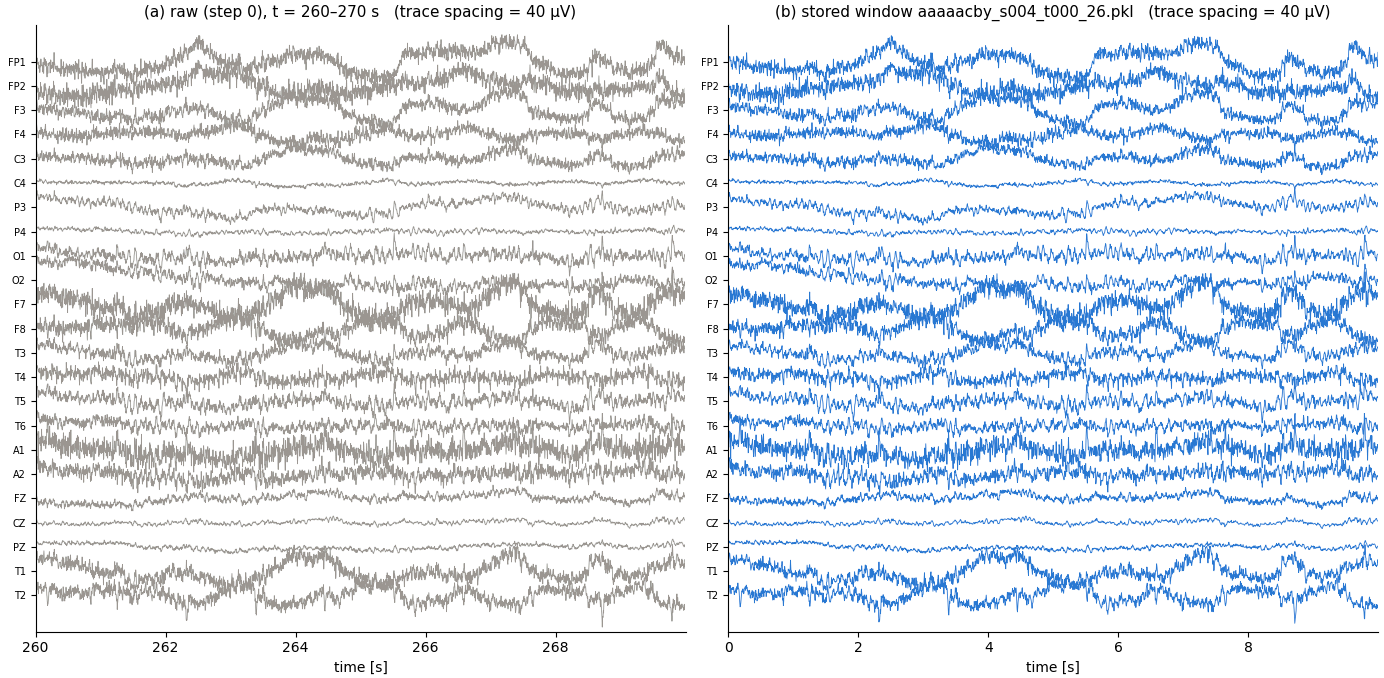


Figure 10.4. Abnormal recording aaaaabdo_s003_t000

121 windows; typical window k* = 40 (Eq. (10.1)): (a) raw, (b) stored.

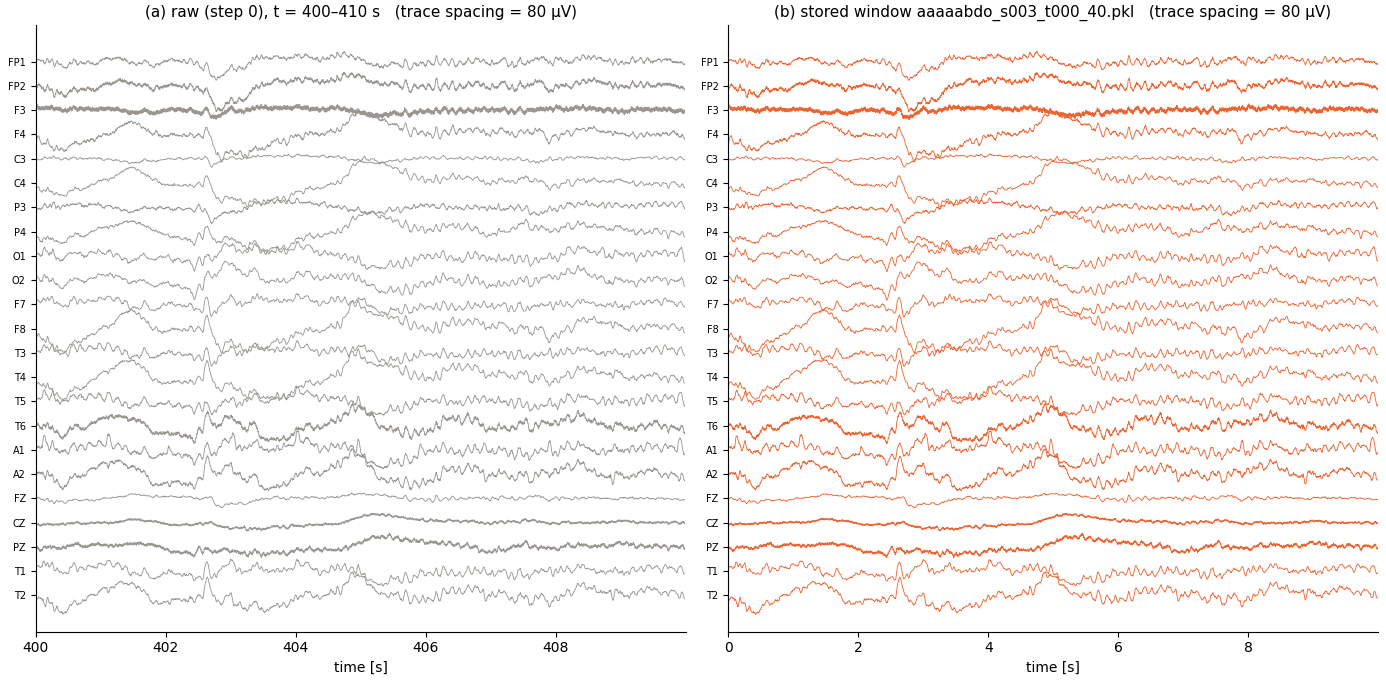


Figure 10.5. Abnormal recording aaaaabsk_s007_t000

142 windows; typical window k* = 91 (Eq. (10.1)): (a) raw, (b) stored.

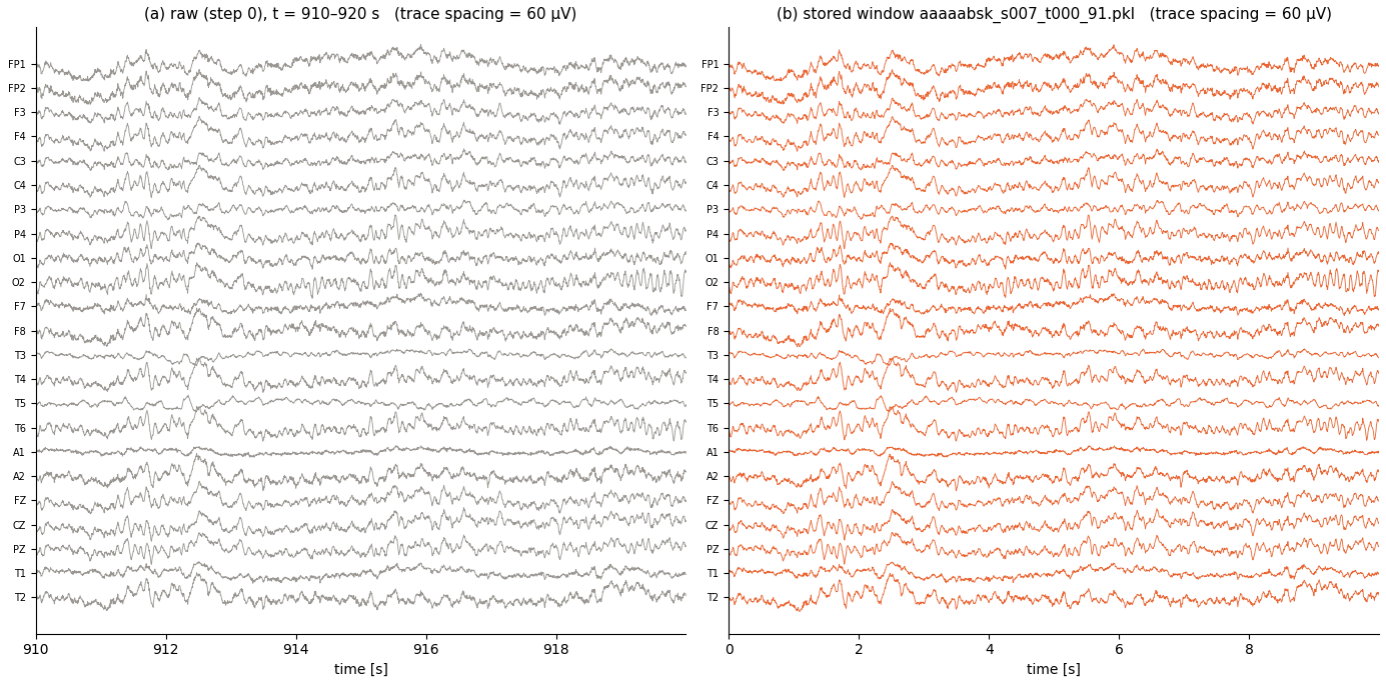
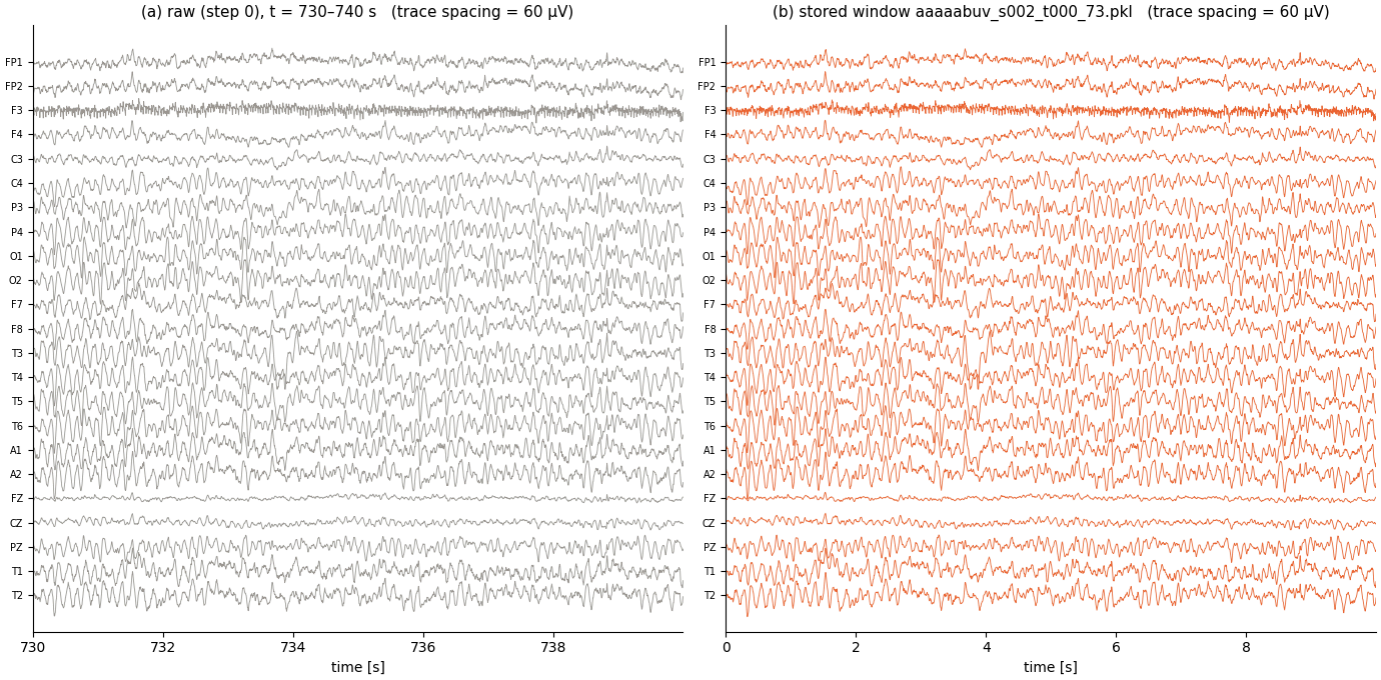


Figure 10.6. Abnormal recording aaaaabuv_s002_t000

138 windows; typical window k* = 73 (Eq. (10.1)): (a) raw, (b) stored.



10.2 Class-level comparison (all test recordings)

A single window says little; Figure 10.7 averages over all test recordings, re-using the per-recording processed spectra $S_r(f)$ of section 2.1 (stored windows, i.e. exactly what the model sees). The **relative band power** of recording r in band b is

$$P_{r,b}^{\mathrm{rel}} = \frac{\int_{f\in b} S_r(f)\,df}{\int_{0.5}^{45} S_r(f)\,df}, \qquad (10.2)$$
$$b \in \{\delta : 0.5\text{–}4,\ \theta : 4\text{–}8,\ \alpha : 8\text{–}13,\ \beta : 13\text{–}30,\ \gamma : 30\text{–}45\}\ \mathrm{Hz}$$

(unitless; the denominator stops at 45 Hz so that the untouched 60 Hz mains line, Table 2.2, does not distort the ratios; integrals by the trapezoidal rule on the 0.25 Hz grid).

Figure 10.7. Normal vs abnormal: spectra of all test recordings

(a) Median processed PSD per class (shaded: inter-quartile range across recordings), all test recordings; (b) relative band power, Eq. (10.2), per recording and class (boxes = quartiles, whiskers = 5th–95th percentile).

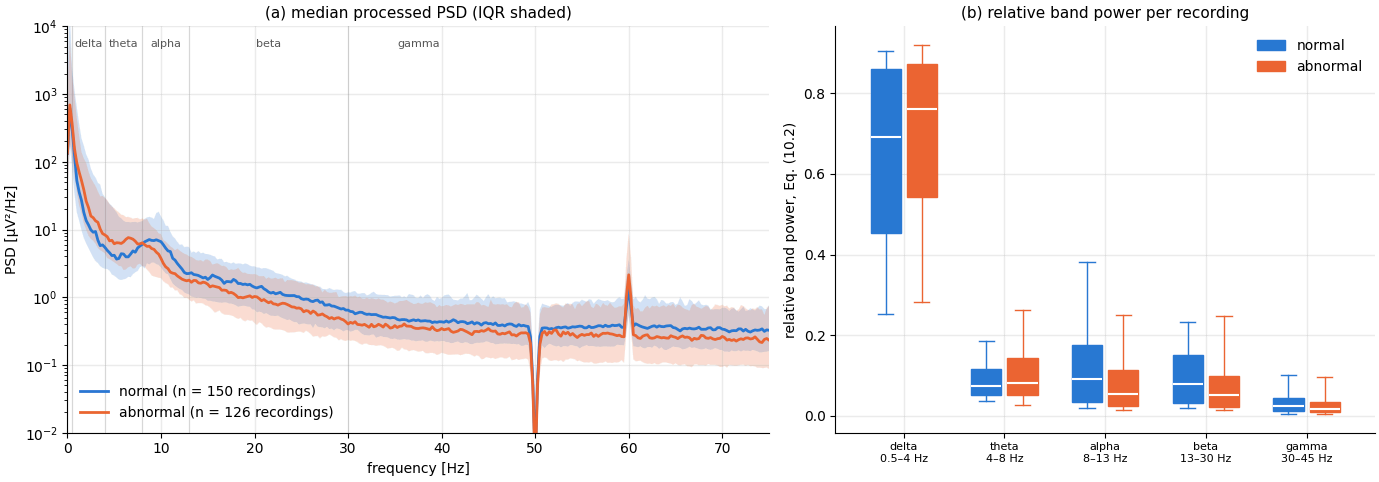


Table 10.1. Relative band power and amplitude per class

Median over test recordings of Eq. (10.2) per band, and of $q_{99}(|X|)$ (µV, the 99th percentile of the absolute amplitude over the sampled stored windows of a recording, as in Eq. (10.1)).

	normal	abnormal	n
P_rel delta	0.692	0.762	150 / 126
P_rel theta	0.072	0.080	150 / 126
P_rel alpha	0.090	0.054	150 / 126
P_rel beta	0.078	0.052	150 / 126
P_rel gamma	0.023	0.017	150 / 126
q99 |X| [µV]	94.212	86.973	150 / 126

Reading Figure 10.7 and Table 10.1. The normal median spectrum has a distinct **alpha peak around 10 Hz**, which is flattened in the abnormal group, and abnormal recordings carry a larger **relative delta and theta** share (slowing). The groups overlap heavily, though — the IQR bands and boxes intersect in every band — which is why the task needs a learned model rather than a single spectral threshold. Both groups show the notch dip at 50 Hz and the untouched mains peak at 60 Hz.

10.3 Spectrogram of whole recordings

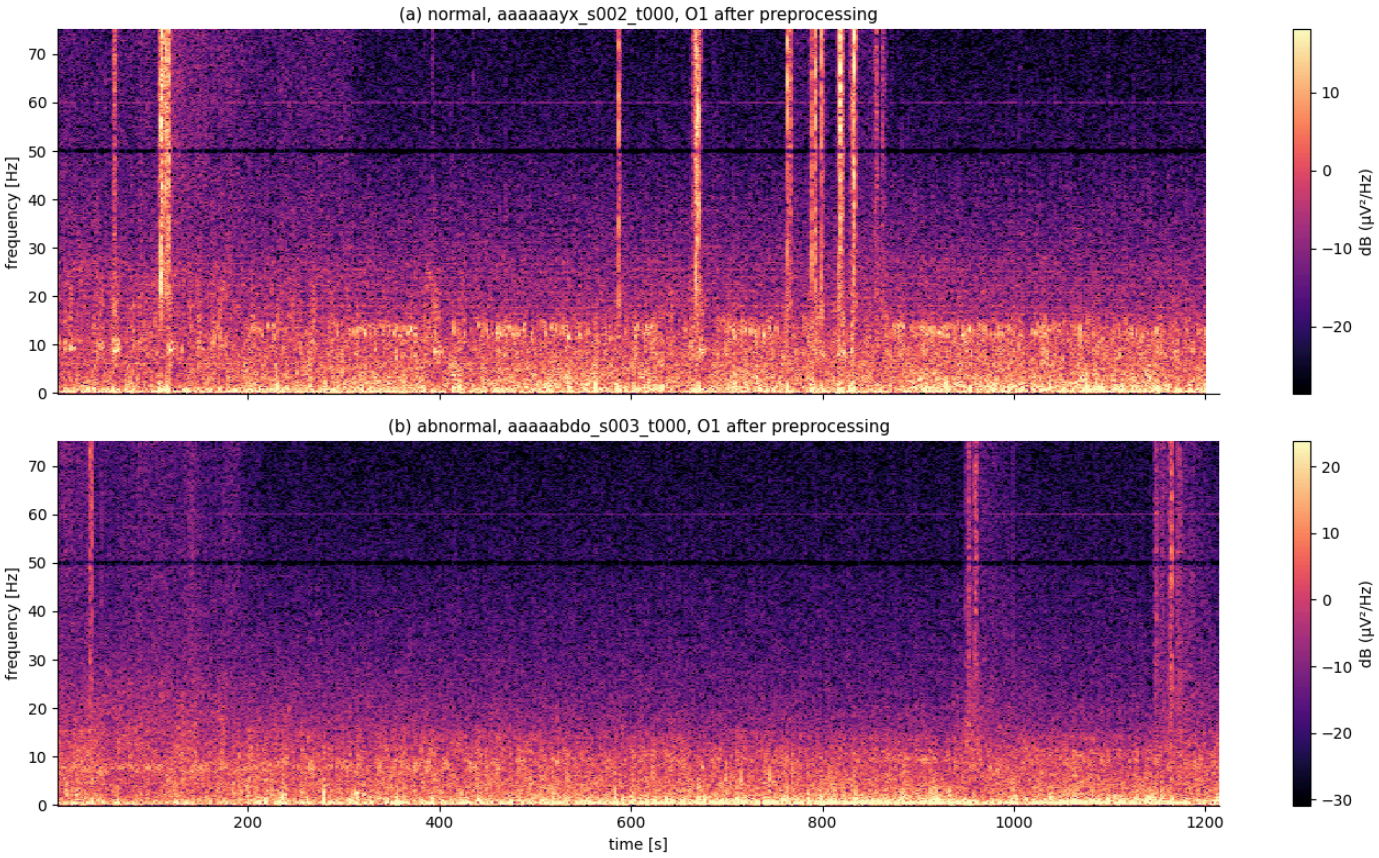
The time-frequency picture over the full recording (after preprocessing, O1 — an occipital channel where the alpha rhythm is strongest), in dB:

$$S(\tau, f) = 10 \log_{10} \frac{2 \left| \sum_{n=0}^{L-1} w[n]\, x[n + \tau L/2]\, e^{-j2\pi f n / f_s} \right|^2}{f_s \sum_n w[n]^2} \qquad (10.3)$$

with a 4 s Tukey window w (`scipy.signal.spectrogram` default), $L = 4 f_s$, 50 % overlap; τ indexes the segment. Horizontal bands = sustained rhythms (alpha ~10 Hz, the faint 60 Hz mains line, the dark 50 Hz notch), vertical stripes = broadband artefacts (movement, electrode pops) that are kept in the training data.

Figure 10.8. Spectrogram of a whole recording

Spectrogram, Eq. (10.3), of O1 after preprocessing (step 4) over the whole recording for (a) the normal and (b) the abnormal example (dB re 1 µV²/Hz).



11. Summary of formulas & caveats

Table 11.1. All steps with their formulas

Step	Operation	Formula	Parameters (MNE defaults in *italics*)
0	EDF → µV	Eq. (2.1)	$f_s = 250$ Hz
1	channel selection	Eq. (3.1)	`drop_channels` , `ch0rder_standard`
2	band-pass	Eqs. (4.1)–(4.5)	0.1–75 Hz, *firwin, 8251 taps, Δf = 0.1 / 18.75 Hz* (Table 4.1)
3	notch	Eq. (5.1)	50 Hz, *width 0.25 Hz, transition 1 Hz*
4	resample	Eqs. (6.1)–(6.2)	250 → 200 Hz, *FFT, boxcar*
5	windowing	Eq. (7.1)	10 s, no overlap, tail dropped
6	model input	Eqs. (8.1)–(8.2)	`[23, 10, 200]` , ÷100
7	*(LaBraM++)*	Eqs. (9.1)–(9.2)	opt-in

Figure 11.1 (end of this section) follows one channel of the same 10 s window through every step of Table 11.1.

Caveats worth knowing

1. **60 Hz mains noise is not removed** (section 5, Table 2.2): the notch is at 50 Hz but TUH is US data.

2. **No artefact handling**: no rejection of bad windows/channels, no re-referencing (unless LaBraM++), only a fixed ÷100 scale. High-amplitude artefact windows go into training as-is (Figure 4.5).
3. **Weak labels**: each 10 s window inherits the recording-level label (Eq. (7.1)); an abnormal recording can contain many normal-looking windows. Recording-level evaluation (aggregating window predictions, see `docs/finetune_evaluation.md`) is closer to the clinical question.
4. **Split reproducibility / leakage**: `make_TUAB.py` calls `np.random.shuffle` **without a seed**, and Eq. (1.1) splits normal and abnormal subjects **independently**, so a subject who has both normal and abnormal recordings can land in train for one and val for the other (Table 11.2). The official `eval/` test set is fixed and unaffected. `docs/age_regression.md` describes a seeded, subject-disjoint split.
5. **Skipped recordings**: files whose channel set does not reduce to exactly the 23 channels are dropped (logged to `tuab-process-error-files.txt`) — counted in Table 11.2.

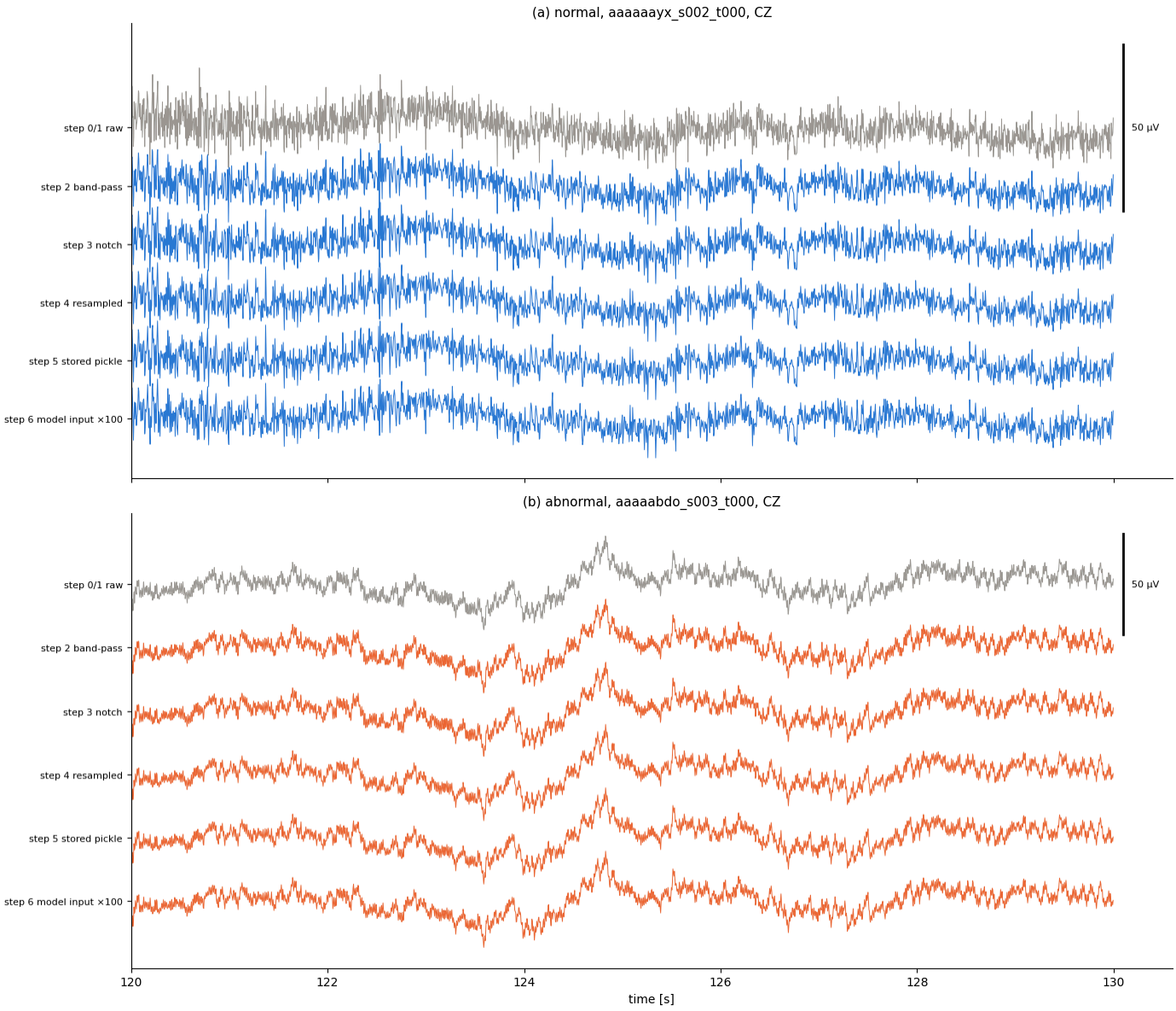
Table 11.2. Split and coverage checks of the stored `processed/` folder

Recording and subject counts per stored split, EDF recordings missing from `processed/` (skipped by Eq. (3.1)), and subject overlaps between splits.

check	value
processed/train recordings / subjects	2142 / 1668
processed/val recordings / subjects	572 / 423
processed/test recordings / subjects	276 / 253
EDF recordings / processed recordings	2993 / 2990
skipped by make_TUAB.py	aaaaajsl_s010_t000, aaaaakfo_s004_t000, aaaaakfo_s005_t000
subjects with normal AND abnormal recordings in train/	54
subjects in both processed/train and processed/val (leakage)	16
subjects shared between train/val and test	0

Figure 11.1. The same 10 s of Cz after every processing step

The **same 10 s window** (`SHOW_T0` … `SHOW_T0 + 10` s, = stored window k = `SHOW_T0`/10) of Cz after every step, for (a) the normal and (b) the abnormal example: step 0/1 raw (250 Hz; channel selection does not change Cz), step 2 band-pass, step 3 notch, step 4 resampled (200 Hz), step 5 stored pickle, step 6 model input (Eq. (8.1), shown ×100). Traces offset vertically, all in µV; scale bar 50 µV.



12. How do other state-of-the-art models prepare EEG?

LaBraM's recipe is one choice among many. Tables 12.1 and 12.2 compare the published preprocessing of **EEG foundation models** and of **EEG age-prediction ("brain age") models**, the downstream task this repo also supports (`data.dataset=TUAB_AGE` , `docs/age_regression.md`). The values were taken from each paper's methods/appendix and, where marked *(code)*, from the official preprocessing scripts. "n/s" = not stated in the paper.

Table 12.1. Preprocessing of EEG foundation models

Model (venue)	Pre-training data	Channels / reference	Band-pass · notch	Rate	Amplitude scaling	Artefact handling	Window · token	Input representation
BENDR (Kostas et al., *Front. Hum. Neurosci.* 2021)	TUEG v1.1–1.2, ~1.5 TB, >10k people	19 ch (10-20, DN3 mapping), missing → 0, **+1 relative-amplitude channel**; no re-ref	none (only Nyquist-safe files kept)	256 Hz	each sequence min–max scaled to **[−1, 1]**	none	60 s · conv tokens ≈ 0.375 s (96× downsampling)	raw time series

Model (venue)	Pre-training data	Channels / reference	Band-pass · notch	Rate	Amplitude scaling	Artefact handling	Window · token	Input representation
BIOT (Yang et al., *NeurIPS* 2023)	SHHS + PREST (private), ~10 M samples; some variants on TUAB/IIIC/CHB-MIT	**16 bipolar TCP** (double banana) computed from `-REF` chans *(code)*; missing channels → dropped tokens	**none** *(code)*	200 Hz	per channel ÷ **95th percentile of |x|** per sample *(code)*, Eq. (12.1)	none	TUAB 10 s · 1 s tokens, 0.5 s overlap	**STFT** energy per token (n_fft = 200, hop 100)
LaBraM (Jiang et al., *ICLR* 2024) — *this repo*	~2,535 h, ~20 datasets (TUSZ, TUEP, SEED, ...)	up to 128 ch, learned 10-20 spatial embedding; TUAB **23 referential** `-REF` chans; no re-ref	**0.1–75 Hz · 50 Hz**	200 Hz	µV ÷ **100** (0.1 mV units), Eq. (8.1)	none	TUAB 10 s · 1 s patches (200 samples), ≤ 256 patches	raw patches (VQ tokenizer target: Fourier amplitude & phase)
EEGPT (Wang et al., *NeurIPS* 2024)	PhysioMI, HGD, TSU, SEED, M3CV	58 ch, **average reference** in pre-training; downstream 1×1-conv "adaptive spatial filter"	n/s for pre-training (0–38 Hz for MI downstream)	256 Hz	**mV** (V × 1000)	n/s	4 s · 250 ms patches (64 samples)	raw patches
CBraMod (Wang et al., *ICLR* 2025)	TUEG, >9,000 h after cleaning	19 ch referential (`-REF` / `-LE`); conv positional encoding; TUAB **16 bipolar TCP** *(code)*	**0.3–75 Hz · 60 Hz**	200 Hz	µV ÷ **100**	drop recordings ≤ 5 min, **drop first & last minute**, **reject 30 s samples with |x| > 100 µV**	30 s (TUAB 10 s) · 1 s patches	raw patch (time-conv) + FFT branch
NeuroLM (Jiang et al., *ICLR* 2025)	~25,000 h (~24,000 h TUEG)	variable, LaBraM-style spatial embedding; TUAB 23 ch	0.1–75 Hz · **50 or 60 Hz by region**	200 Hz	µV ÷ **100**	n/s	TUAB 10 s · 1 s patches, ≤ 1,024 patches	VQ tokens → GPT-2 (multi-task LLM)
Neuro-GPT (Cui et al., *ISBI* 2024)	TUH EEG, 20k recordings, 5,656 h	22 ch, bad chans interpolated, **average reference**	**0.5–100 Hz · 60 Hz**	250 Hz	DC removal, detrend, **z-score per recording**	bad-channel interpolation	32 × 2 s chunks (10 % overlap)	raw chunks → conv + transformer → GPT-2
FEMBA (Tegon et al., arXiv 2025)	TUEG ~21,000 h (TUAB/TUAR/TUSL removed)	n/s	n/s	n/s	per channel **IQR (quartile) scaling**	n/s	2-D patches (4 ch × 32 samples)	raw patches → bidirectional Mamba
REVE (El Ouahidi et al., *NeurIPS* 2025)	61,415 h, 92 datasets, ~25k subjects	any montage, **4-D (xyz + time) positional encoding**	**0.5–99.5 Hz** · n/s	200 Hz	**z-score** (session stats), **clip ±15 SD**	none (no rejection)	~10 s · 1 s patches (0.1 s overlap)	raw patches

Footnotes: BENDR/BIOT/EEGPT do not report pre-training hours. EEGPT's paper text says 256 Hz for TUAB, but its released TUAB script is LaBraM's pipeline (0.1–75 Hz, 50 Hz notch, 200 Hz, ÷100). FEMBA's filter/rate/montage are not given in the paper.

Table 12.2. Preprocessing of EEG age-prediction (brain-age) models

Work	Data (N, age)	Channels / reference	Filtering	Rate	Artefact handling	Segments	Scaling	Features → model	Recording-level age	MAE
Engemann et al., *NeuroImage* 2022 — M/EEG brain-age benchmark *(values from code)*	TUAB **normal** recordings, 1,385 subj., 0–95 y (+ Cam-CAN, LEMON, CHBP)	21 ch (10-20 + A1/A2), **average reference**	**0.1–49 Hz**, no notch	200 Hz	**autoreject** on epochs	10 s	µV ÷ 9.7 (dataset SD) for deep nets, Eq. (12.2)	filter-bank **Riemann** covariances (7 bands) + ridge; hand-crafted + RF; ShallowNet / Deep4	mean of window predictions	TUAB: Deep4 7.75, Shallow 7.80, Riemann 8.25 y
Gemein et al., *Imaging Neurosci.* 2024 — "Brain age revisited"	TUAB, 2,647 rec. / 2,159 subj. (train on non-pathological)	21 ch, **common average**	n/s	**100 Hz**	**drop first minute**, **clip ±800 µV**	braindecode cropped decoding (stride 1)	channel z-score (train stats), Eq. (12.3); target min-max [0, 1]	BD-TCN, L1 loss	crop mean (likely; not explicit)	6.6 y (R^2 0.73)
Gemein et al., *NeuroImage* 2020 — TUAB **pathology** (the reference braindecode TUAB recipe)	TUAB 2,993 rec.	21 ch	none	100 Hz	drop first 60 s, ≤ 20 min, clip ±800 µV	~6 s receptive-field crops	—	Deep4 / Shallow / TCN / EEGNet; RF / SVM / Riemann	crop aggregation	(classification, 86 % acc.)
Sun et al., *Neurobiol. Aging* 2019 — sleep EEG	MGH PSG, 2,532 (18–80 y)	6 ch (F3/F4/C3/C4/O1/O2 – contralateral mastoid)	**0.5–20 Hz · 60 Hz**	200 Hz	reject 30 s epochs > 500 µV, flat > 2 s, ECG-contaminated	30 s sleep epochs	per channel median / IQR over the night	102 features × 5 sleep stages → regularised linear model	features averaged per stage	7.8 y
Brink-Kjaer et al., *npj Digit. Med.* 2022 — deep PSG	13,332 PSGs, 7 cohorts, 6–90 y	C3-M2, C4-M1 (+ EOG/EMG/ECG variants)	0.3–45 Hz elliptic IIR, zero phase	128 Hz	n/s	5 min epochs	5th/95th percentile → [−1, 1]	CNN + Bi-LSTM + attention, Huber loss	2nd-stage Bi-LSTM + attention over the night	5.8 y
Khayretdinova et al., *Front. Aging Neurosci.* 2022	TD-BRAIN, 1,335 sessions, 5–88 y	26 ch (10-10), averaged mastoids	**0.5–100 Hz · 50/60 Hz**	500 Hz	EOG regression + automatic EMG / jump / kur-	5 s, 1 s step	n/s	deep CNN with channel-permutation stacking, L1 loss	mean of clean-segment predictions	5.96 y

Work	Data (N, age)	Channels / reference	Filtering	Rate	Artefact handling	Segments	Scaling	Features → model	Recording-level age	MAE
					tosis / blink detection, flagged segments dropped					
Sabbagh et al., *BJA Open* 2023 — anaesthesia EEG	323 patients ≥ 18 y	4 frontal (Fp1/Fp2/F7/F8 – Cz)	n/s	63 Hz	burst-suppression / artefacts removed by vendor export; keep stable-anaesthesia segment	60 s, 10 s shift	—	PSD + 5-band covariances (coffeine) → ridge	averaged over epochs	~8.2 y
Banville et al., *Imaging Neurosci.* 2024 — Muse mobile EEG	4,191 meditation + 1,020 sleep rec., 18–81 y	4 ch (TP9, Fp1, Fp2, TP10), Fpz ref	0.1–49 Hz FIR + notches 16 … 60 Hz	128 Hz	crop first minute; reject windows with peak-to-peak > 250 µV	10 s / 30 s	—	filter-bank covariances → ridge	averaged features	9.3 y / 7.6 y
This repo, `TUAB_AGE` (`docs/age_regression.md`)	TUAB, age from EDF header, subject-disjoint split	23 ch referential (LaBraM)	0.1–75 Hz · 50 Hz	200 Hz	none	10 s	µV ÷ 100; **target** z-scored with train stats	fine-tuned LaBraM, Huber loss	mean / median / max of windows	— (literature range 7–8 y)

12.3 What differs, and why it matters

- **Montage.** On TUAB, BIOT and CBraMod use the **16-channel bipolar TCP** montage; LaBraM, NeuroLM and EEGPT use **23 referential** channels; the brain-age works use **21 channels re-referenced to the common average**. Bipolar and average-reference montages remove the signal shared by the electrodes in each derivation, e.g. the activity of the shared reference electrode; with referential channels (LaBraM) that shared component stays in every channel.
- **Line noise.** TUH is US data (60 Hz). LaBraM/EEGPT notch at 50 Hz (section 5), CBraMod and Neuro-GPT at 60 Hz, NeuroLM by region; the TUAB age benchmarks avoid the issue with a ≤ 49 Hz low-pass or a 100 Hz rate (Nyquist 50 Hz).
- **Amplitude scaling** falls into three families: a **fixed physical unit** (÷100 µV: LaBraM, CBraMod, NeuroLM; mV: EEGPT; ÷9.7 µV: Engemann) keeps absolute amplitude, which carries information (amplitude falls with age); **per-sample/per-channel normalisation** (BIOT 95th percentile, FEMBA IQR, BENDR min-max) discards it; **z-scoring with dataset/session statistics** (Gemein, REVE, Neuro-GPT) is in between.
- **Artefacts.** The foundation models mostly keep everything (CBraMod is the exception, rejecting windows with |x| > 100 µV and the first/last minute), whereas the brain-age literature almost always removes or clips artefacts (autoreject, ±800 µV clip, peak-to-peak thresholds, first-minute crop).
- **Recording-level aggregation.** Every brain-age work predicts **one age per recording**, typically by averaging window predictions or features, which is why this repo evaluates age per recording.

12.4 Five recipes applied to the same excerpt

The next cells re-implement five of these recipes (simplified, see the code comments) on the step-1 signal x_c (µV) of the abnormal example. Besides the filters and rates of Table 12.1/12.2, they scale amplitudes as

$$\hat{x}_c = \frac{x_c}{q_{95}\big(|x_c|\big) + 10^{-8}} \quad \text{(BIOT, per channel and 10 s sample)} \tag{12.1}$$

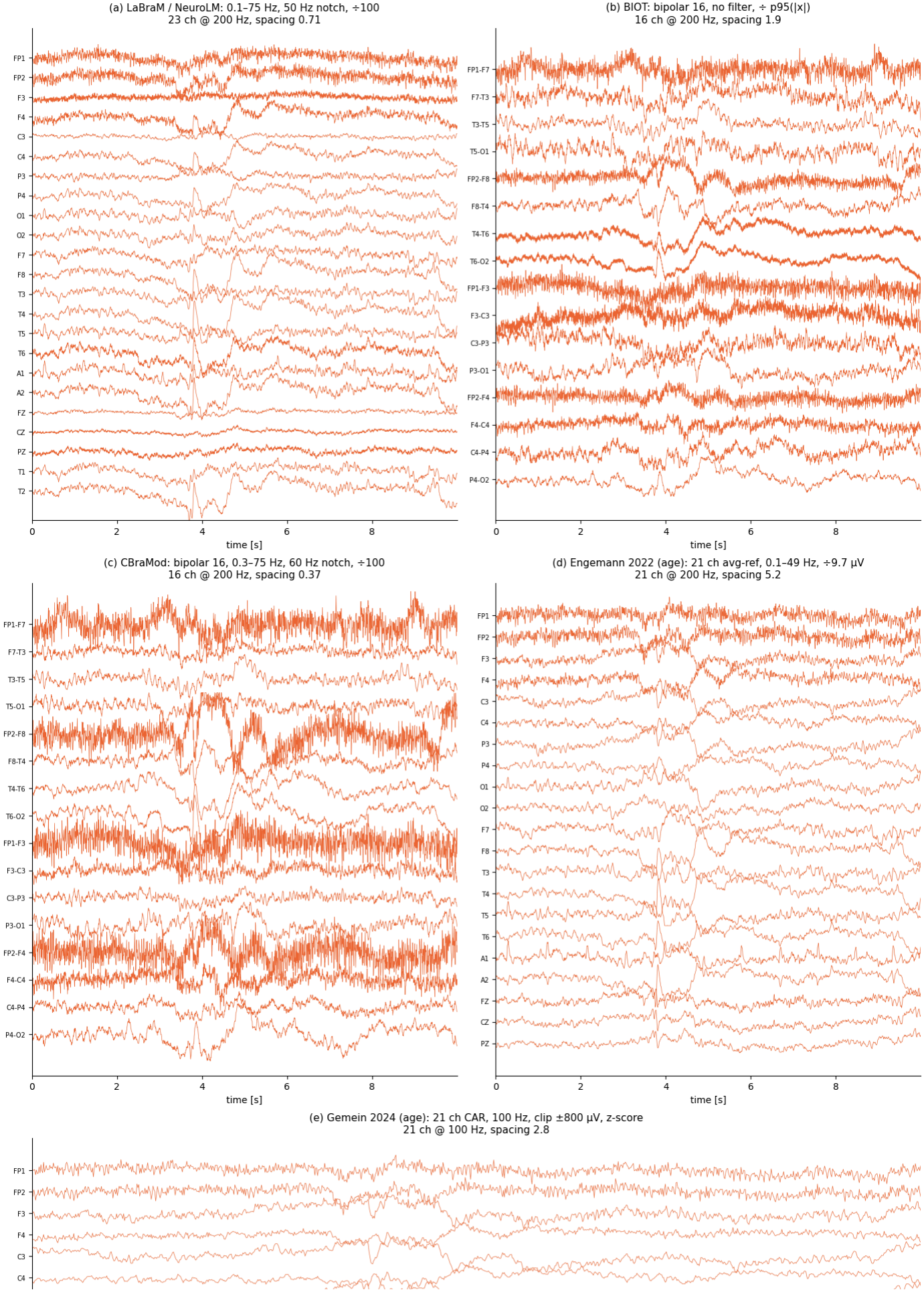
$$\hat{x}_c = \frac{1}{9.7}\Big(x_c - \frac{1}{C}\sum_{c'} x_{c'}\Big) \quad \text{(Engemann 2022: average reference, } \div 9.7\ \mu\text{V)} \tag{12.2}$$

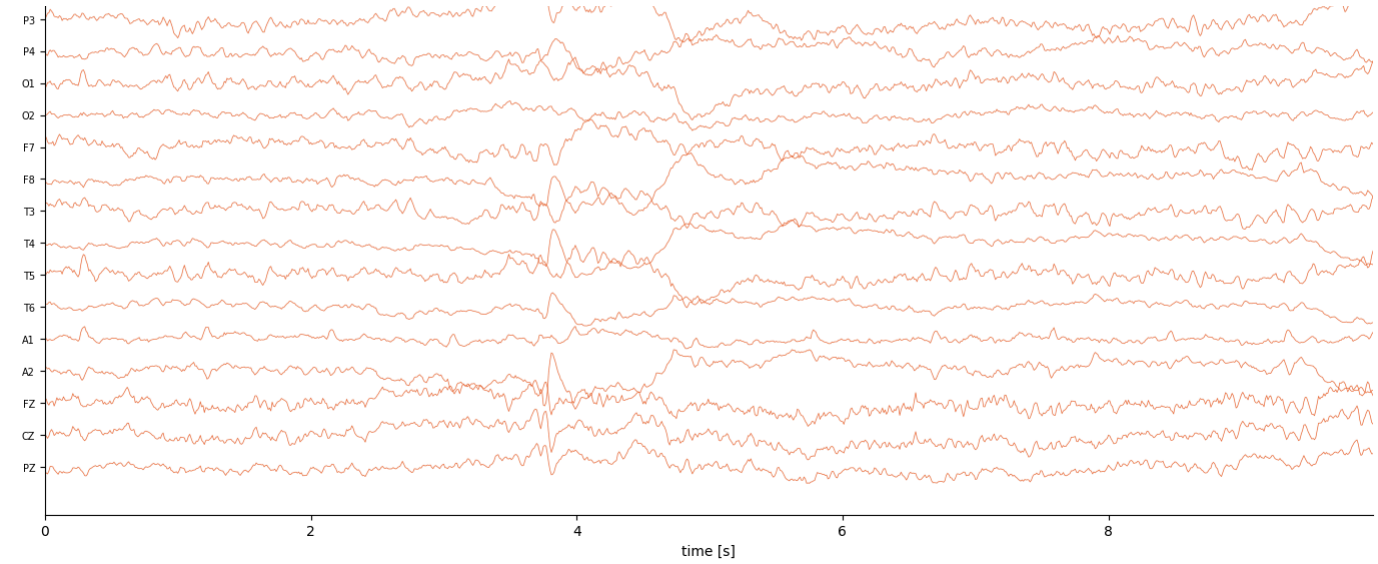
$$\begin{aligned} \hat{x}_c &= \frac{u_c - \operatorname{mean}(u_c)}{\operatorname{std}(u_c)} \quad \text{(Gemein 2024),} \\ u_c &= \operatorname{clip}_{\pm 800}(x_c) - \frac{1}{C}\sum_{c'} \operatorname{clip}_{\pm 800}(x_{c'}) \end{aligned} \tag{12.3}$$

where q_{95} is the 95th percentile over the samples of the window, $C = 21$ channels, and mean/std are over time (first minute excluded; this recording's statistics stand in for the training-set statistics). LaBraM and CBraMod use the ÷100 of Eq. (8.1). Figure 12.1 shows the same 10 s under each recipe.

Figure 12.1. The same abnormal excerpt under five published recipes

The abnormal excerpt (`SHOW_T0` … `SHOW_T0 + 10` s) after (a) LaBraM / NeuroLM, (b) BIOT, (c) CBraMod, (d) Engemann 2022 and (e) Gemein 2024 preprocessing; each panel in that recipe's model units (spacing in the title), bipolar recipes show the 16 TCP derivations.


(a) LaBraM / NeuroLM: 0.1–75 Hz, 50 Hz notch, ÷100
23 ch @ 200 Hz, spacing 0.71
FP1
FP2
F3
F4
C3
C4
P3
P4
O1
O2
F7
F8
T3
T4
T5
T6
A1
A2
FZ
CZ
PZ
T1
T2
0
2
4
6
8
time [s]
(b) BIOT: bipolar 16, no filter, ÷ p95(|x|)
16 ch @ 200 Hz, spacing 1.9
FP1-F7
F7-T3
T3-T5
T5-O1
FP2-F8
F8-T4
T4-T6
T6-O2
FP1-F3
F3-C3
C3-P3
P3-O1
FP2-F4
F4-C4
C4-P4
P4-O2
0
2
4
6
8
time [s]
(c) CBraMod: bipolar 16, 0.3–75 Hz, 60 Hz notch, ÷100
16 ch @ 200 Hz, spacing 0.37
FP1-F7
F7-T3
T3-T5
T5-O1
FP2-F8
F8-T4
T4-T6
T6-O2
FP1-F3
F3-C3
C3-P3
P3-O1
FP2-F4
F4-C4
C4-P4
P4-O2
0
2
4
6
8
time [s]
(d) Engemann 2022 (age): 21 ch avg-ref, 0.1–49 Hz, ÷9.7 µV
21 ch @ 200 Hz, spacing 5.2
FP1
FP2
F3
F4
C3
C4
P3
P4
O1
O2
F7
F8
T3
T4
T5
T6
A1
A2
FZ
CZ
PZ
0
2
4
6
8
time [s]
(e) Gemein 2024 (age): 21 ch CAR, 100 Hz, clip ±800 µV, z-score
21 ch @ 100 Hz, spacing 2.8
FP1
FP2
F3
F4
C3
C4



Which frequencies does each recipe pass to the model? Figure 12.2 shows the spectrum of the whole recording under each recipe, normalised to its own total power so that the different units do not matter:

$$\hat{S}(f) = \frac{S(f)}{\int_0^{f_s/2} S(f')\,df'} \tag{12.4}$$

where S is Eq. (2.3) of the recipe's output (unit 1/Hz). On this recording only CBraMod's 60 Hz notch, Engemann's 49 Hz low-pass and Gemein's 100 Hz rate keep the mains line out of the model input; the re-montaging alone (bipolar in BIOT) does not remove it (Table 12.3).

Figure 12.2. Spectrum passed to the model under each recipe

Normalised spectrum, Eq. (12.4), of the abnormal example under each recipe of Figure 12.1 (dotted: 50 and 60 Hz).

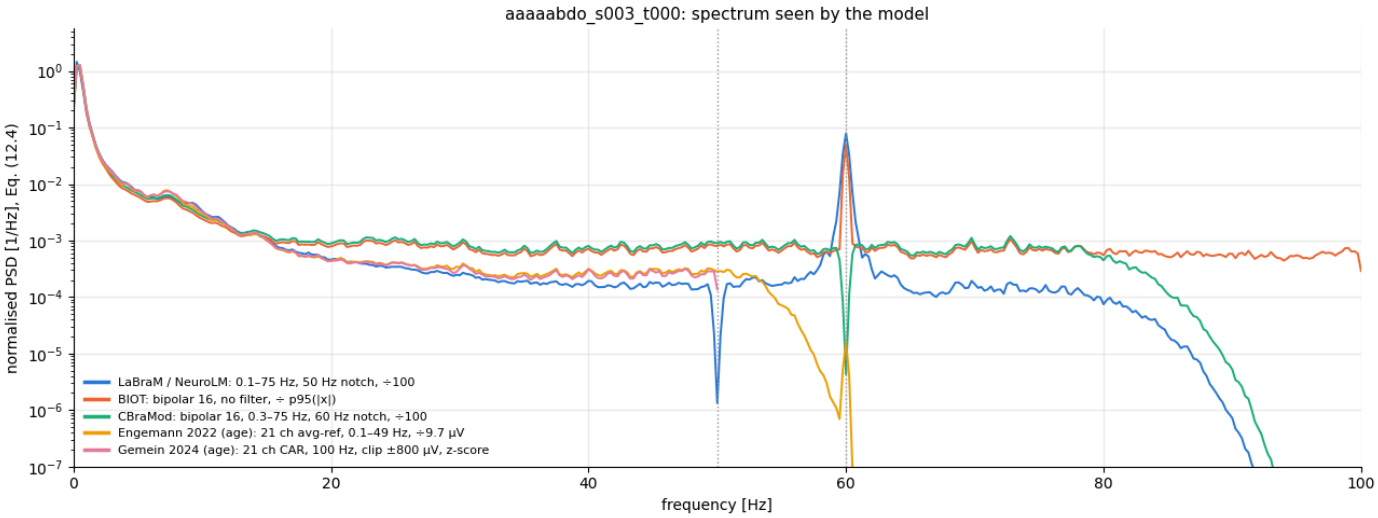


Table 12.3. Mains-line power reaching the model under each recipe

$\rho(60\ \mathrm{Hz})$ of Eq. (2.6) in dB for the abnormal example (8 s Welch segments); "none" when the recipe's Nyquist frequency is below 60 Hz.

recipe	ρ(60 Hz) [dB], Eq. (2.6)
LaBraM / NeuroLM: 0.1–75 Hz, 50 Hz notch, ÷100	-6.2
BIOT: bipolar 16, no filter, ÷ p95(|x|)	-9.4
CBraMod: bipolar 16, 0.3–75 Hz, 60 Hz notch, ÷100	-31.5
Engemann 2022 (age): 21 ch avg-ref, 0.1–49 Hz, ÷9.7 µV	-44.2
Gemein 2024 (age): 21 ch CAR, 100 Hz, clip ±800 µV, z-score	none (Nyquist 50 Hz)

References (preprocessing values verified against the papers' methods sections and, where noted, official code)

- Kostas, Aroca-Ouellette & Rudzicz (2021). *BENDR.* Front. Hum. Neurosci. — arXiv:2101.12037
- Yang, Westover & Sun (2023). *BIOT.* NeurIPS — arXiv:2305.10351, code: github.com/ycq091044/BIOT
- Jiang, Zhao & Lu (2024). *LaBraM.* ICLR — arXiv:2405.18765, code: github.com/935963004/LaBraM
- Wang et al. (2024). *EEGPT.* NeurIPS — paper, code: github.com/BINE022/EEGPT
- Wang et al. (2025). *CBraMod.* ICLR — arXiv:2412.07236, code: github.com/wjq-learning/CBraMod
- Jiang et al. (2025). *NeuroLM.* ICLR — arXiv:2409.00101
- Cui et al. (2024). *Neuro-GPT.* ISBI — arXiv:2311.03764
- Tegon et al. (2025). *FEMBA.* — arXiv:2502.06438
- El Ouahidi et al. (2025). *REVE.* NeurIPS — arXiv:2510.21585
- Engemann et al. (2022). *A reusable benchmark of brain-age prediction from M/EEG resting-state signals.* NeuroImage 262:119521 — doi, code: github.com/meeg-ml-benchmarks/brain-age-benchmark-paper
- Gemein et al. (2024). *Brain age revisited.* Imaging Neuroscience — doi:10.1162/imag_a_00210
- Gemein et al. (2020). *Machine-learning-based diagnostics of EEG pathology.* NeuroImage 220:117021 — arXiv:2002.05115
- Sun et al. (2019). *Brain age from the electroencephalogram of sleep.* Neurobiol. Aging 74:112–120 — doi
- Brink-Kjaer et al. (2022). *Age estimation from sleep studies using deep learning.* npj Digit. Med. 5:103 — doi
- Khayretdinova et al. (2022). *Predicting age from resting-state scalp EEG with deep CNNs on TD-BRAIN.* Front. Aging Neurosci. 14:1019869 — doi
- Sabbagh et al. (2023). *Repurposing EEG monitoring of general anaesthesia for brain-age prediction.* BJA Open — doi
- Banville et al. (2024). *Brain age from mobile (Muse) EEG.* Imaging Neuroscience — doi

Where to go next

- `dataset_maker/make_TUAB.py` — the offline script (run it once to create `processed/`).
- `labram/data/tuh_datasets.py` — `TUABLoader` / `prepare_TUAB_dataset` (reads the pickles).
- `labram/train/train_finetune.py` — ÷100 scaling and patching.
- `docs/labram_plus_plus.md` — CAR + z-score + sin/cos phase loss.
- `notebooks/finetune_evaluation.ipynb` — evaluating a fine-tuned model per window and per recording.

13. Export to PDF

The PDF next to this notebook (`tuab_data_preparation.pdf`) is rendered from the **saved** notebook by `scripts/notebook_to_pdf.py`: nbconvert's `webpdf` exporter prints the HTML view with headless Chromium, so MathJax typesets every `$$ … \tag{s.k} $$` formula with its number, Markdown and pandas tables and figures keep their `Table s.k` / `Figure s.k` headings. Code cells are hidden (`--no-input`); cells tagged `pdf-skip` (this section's code) are dropped.

```
pip install "nbconvert[webpdf]" && python -m playwright install chromium   # once
python scripts/notebook_to_pdf.py notebooks/tuab_data_preparation.ipynb --execute
```

Set `EXPORT_PDF = True` and run the cell below to export from here (save the notebook first: the exporter reads the file on disk); the next cell shows the PDF inline.